Lecture on vision sensor/devices inspired by natural eye functions

The University of Electro-Communications
Associate Professor
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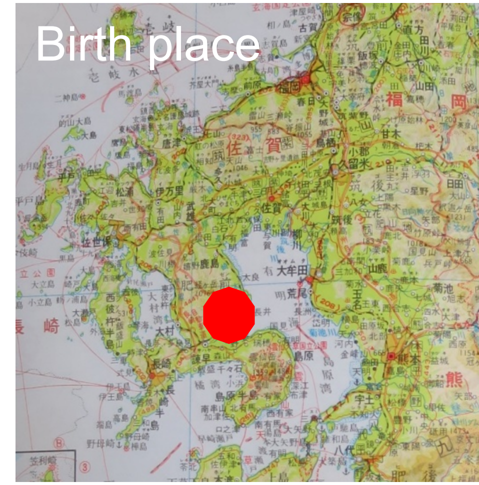
Keio Univerisity 2021/06/14

Self Introduction

- Born 1978/July/24 at Nagasaki pref.



- Education

2001 The Univ. of Tokyo, Faculty of Eng.

2003 The Univ. of Tokyo, Graduate School of Information Science, Mater

2006 The Univ. of Tokyo, Graduate School of Information Science, Ph.D.

- Job

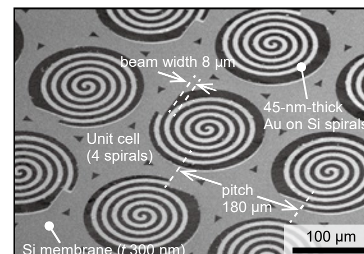
2006~ PosDoc Researcher at UT
2008~2015 Assistant Professor at UT

2015~2016 Lecturer at UT

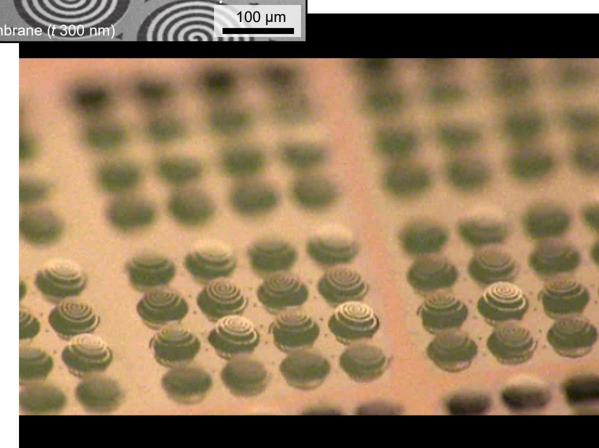
2016~ Associate Professor at UEC

- Majoring

Optical Micro Sensors and Devices using MEMS (Micro Electro Mechanical Systems) technology



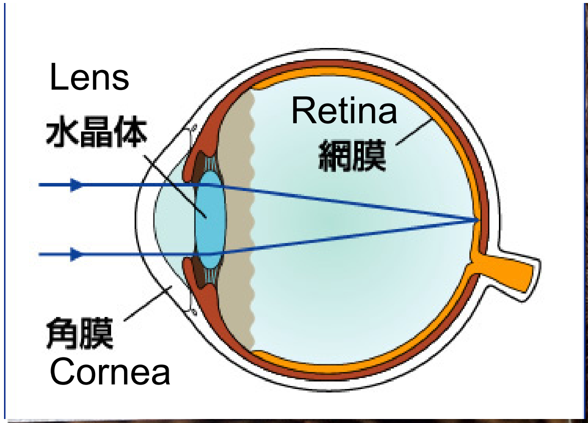
THz MEMS metamaterial

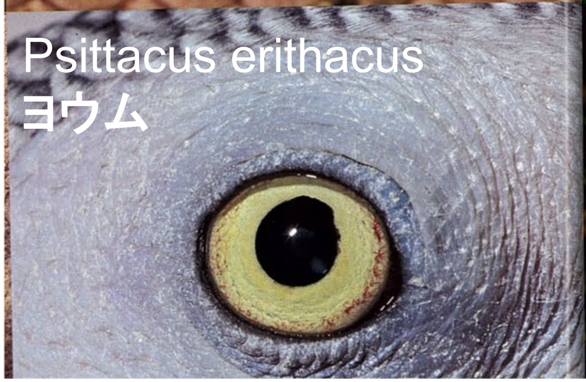


Contents

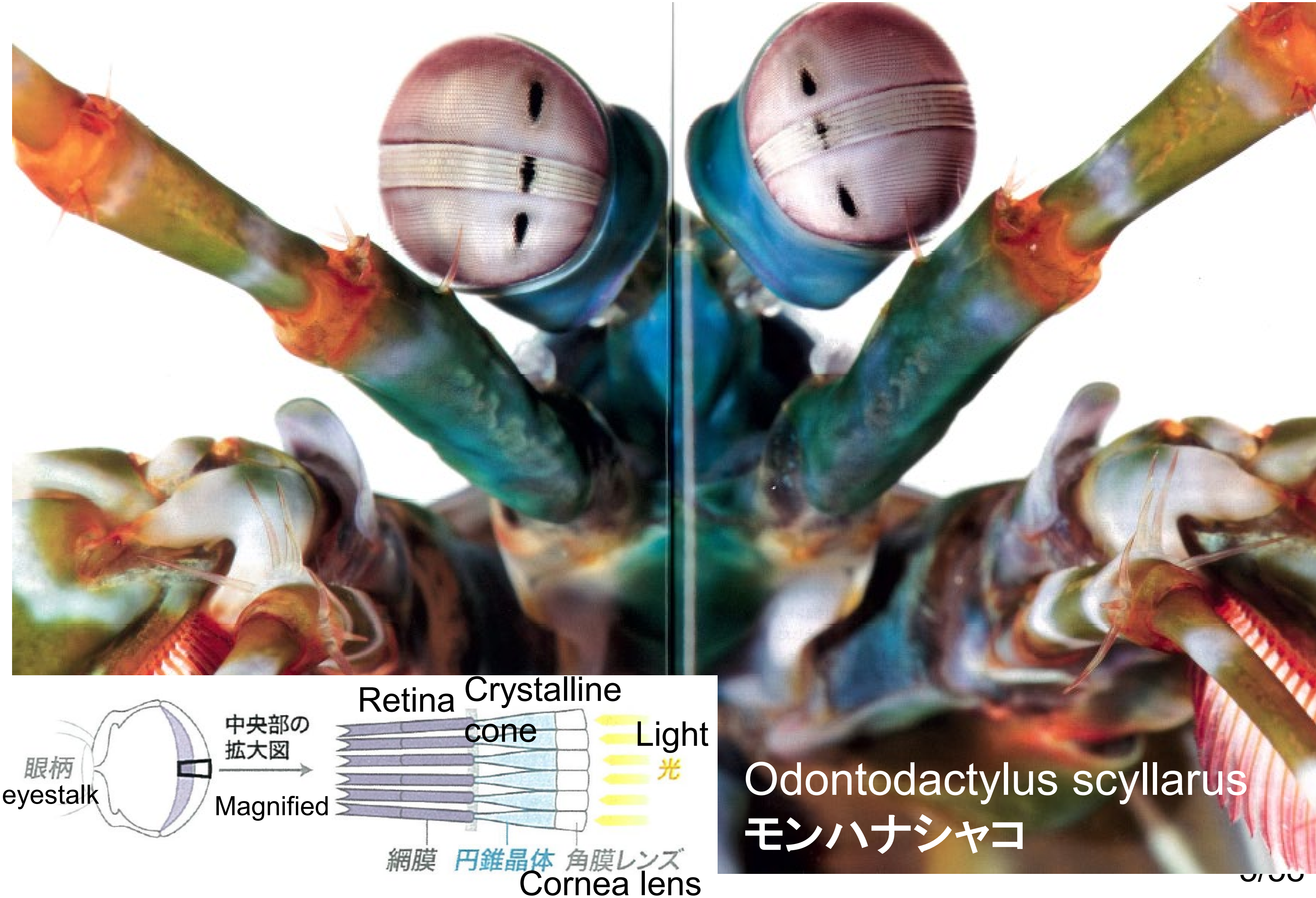
Eyes of vertebrate animals

Architecture is the same

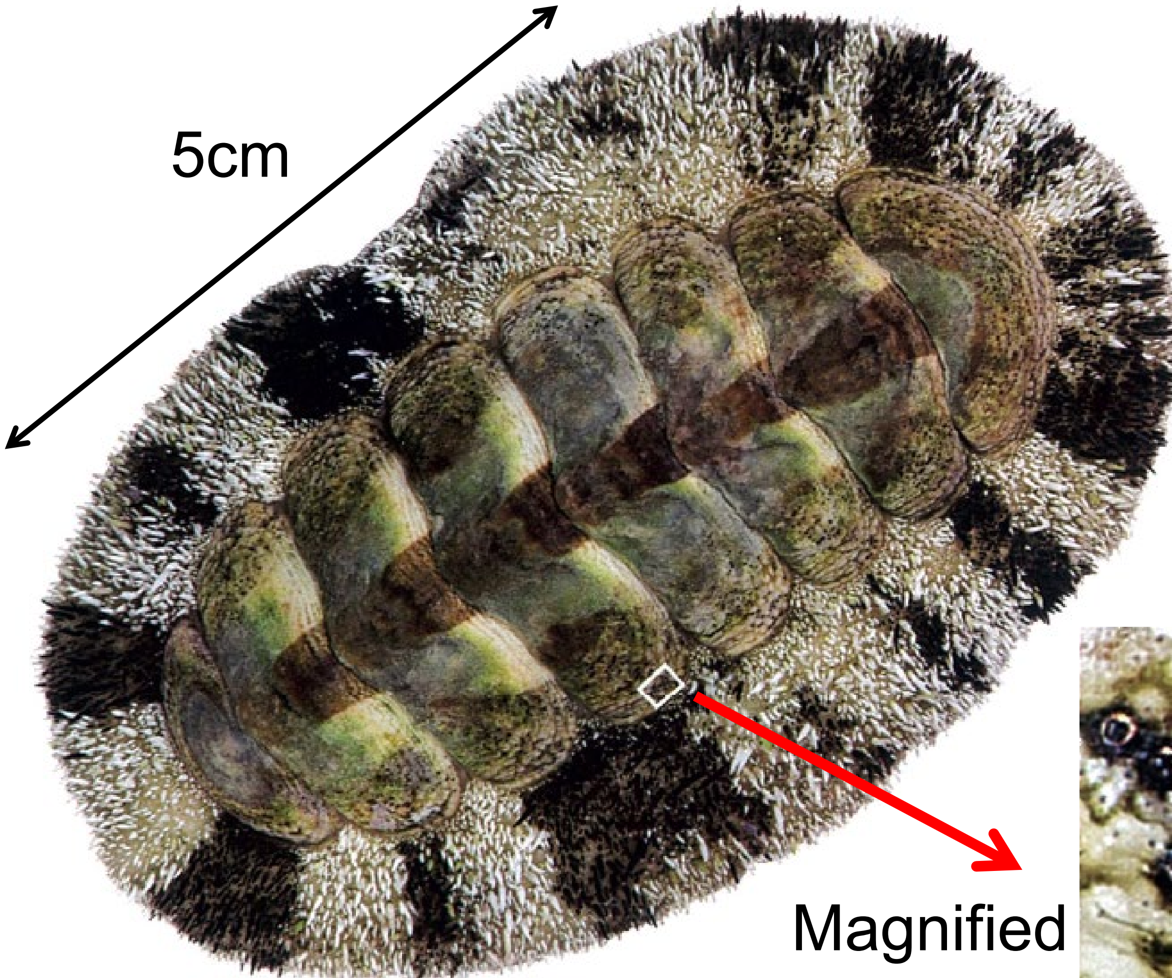




Other architecture of eye (Crustacea)



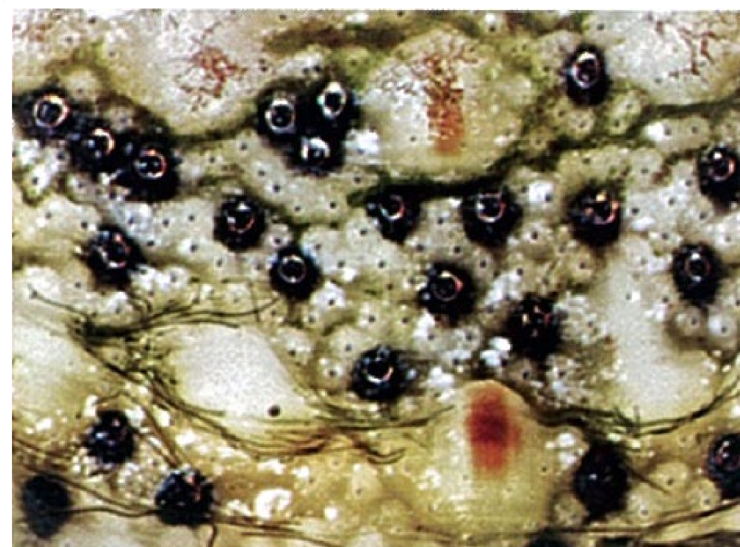
Other architecture of eye (shellfish)



ウニヒザラガイ
Chiton

Black spots = eyes with lens, retina, pigment cell
Lens is made of crystalline $CaCO_3$.

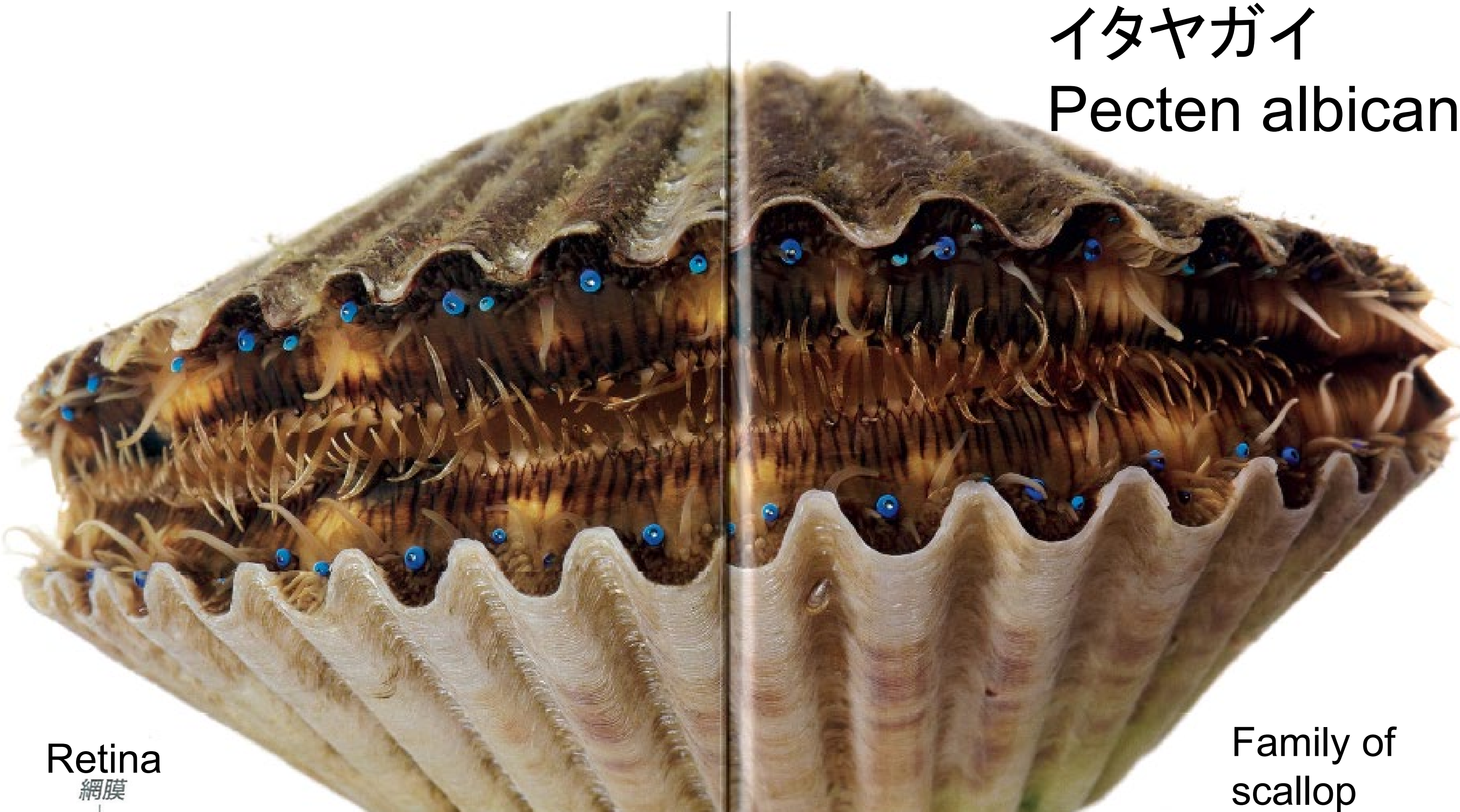
Magnified

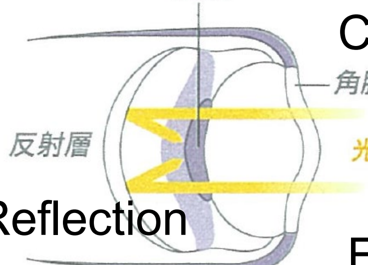


Other architecture of eye (shellfish)

イタヤガイ

Pecten albicans





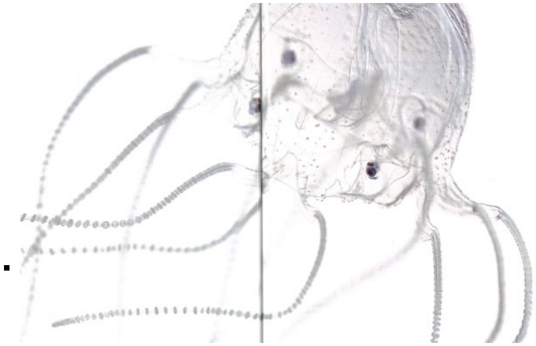
Family of scallop

Eyes on a mantile.

Eyes are reflective lens (Cassegrain type).

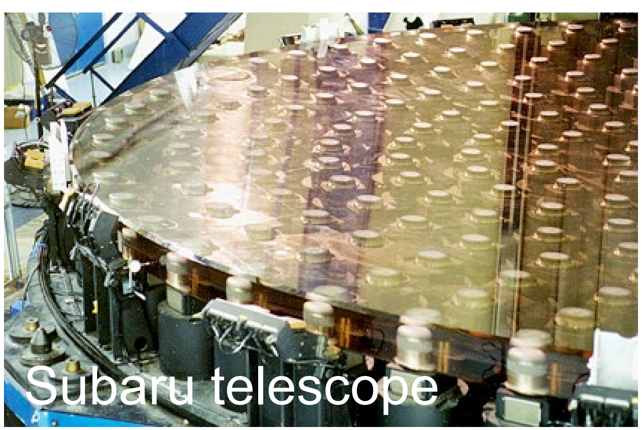
Image resolution gives different world views

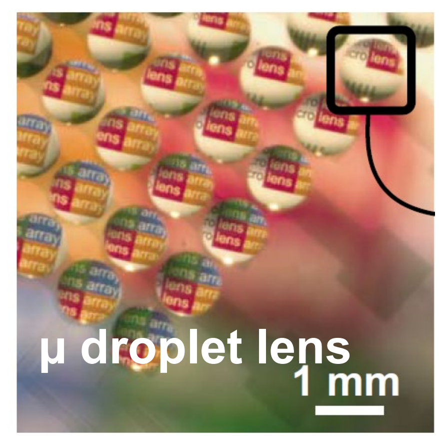
Jellyfish
Some eyes sees water surface.
Not good resolution.



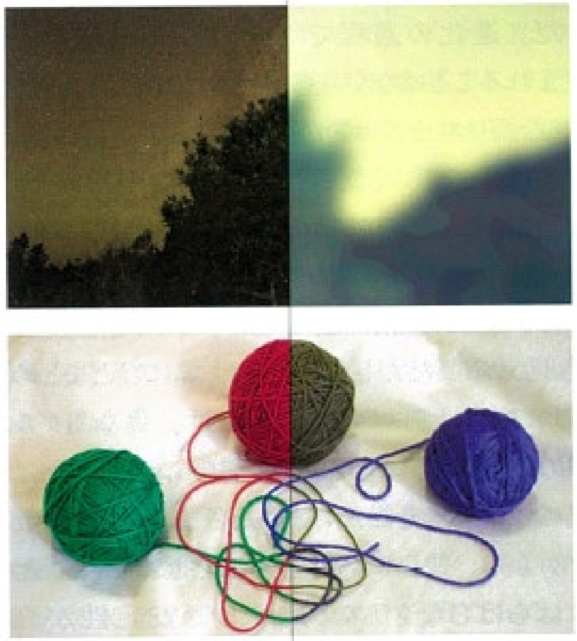
Eagle
Two times better resolution than human's eye.
High pixel density in retina.

Artificial Eyes
Resolution selected depending on usage





Photodetector sensitivity provides different world view



スズメガ

Deilephila elpenor

夜行性のスズメガの一種、ベニスズメの目は、光に対する感受性が高い。このため月が出ていない夜、星明かりだけでも色を識別できる。おかげで夜間でも、色を手がかりに花を見つけて花蜜にありつける。人間の目にはできない芸当だ。

ネコ

Felis catus

イエネコの目には、かすかな光も感知する桿体細胞(かんたい)が人間の目よりもたくさんある。さらに、縦長の瞳孔が暗がりでは大きく開くため、夜間に狩りをするのに適している。色を見分ける錐体細胞は少ないため、緑と赤を見分けられない。

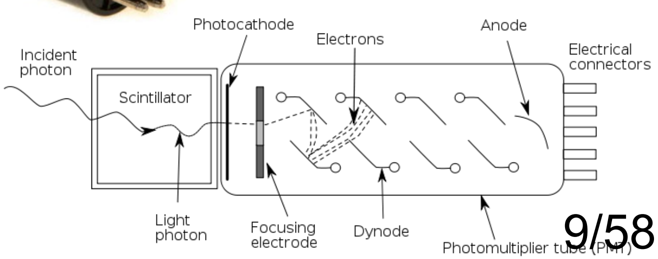
Hawkmoth

Photodetector sensitivity difference presents different sensation.

Adaptive sensitivity selection by revolution.

Highly sensitive Photodetector (PMT) by Hamamatsu Photonics

We also change sensors depending on required sensitivity.



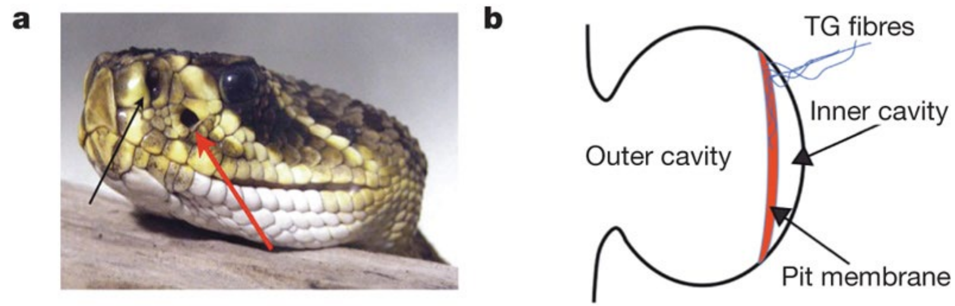
Different wavelength sensitivity provides different world view

Bees can see UV.

Center Highlighted image of flower

Bees and plans make use of wave-length other than visible

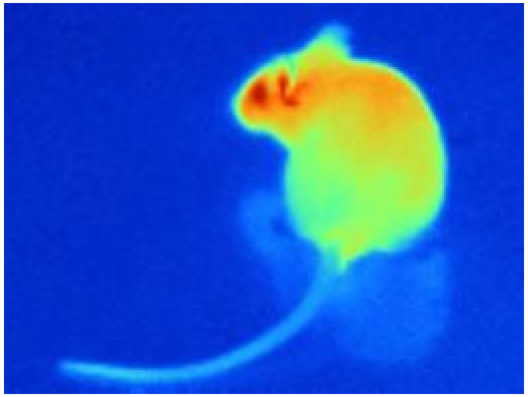
http://www.visualnews.com



doi:10.1038/nature08943

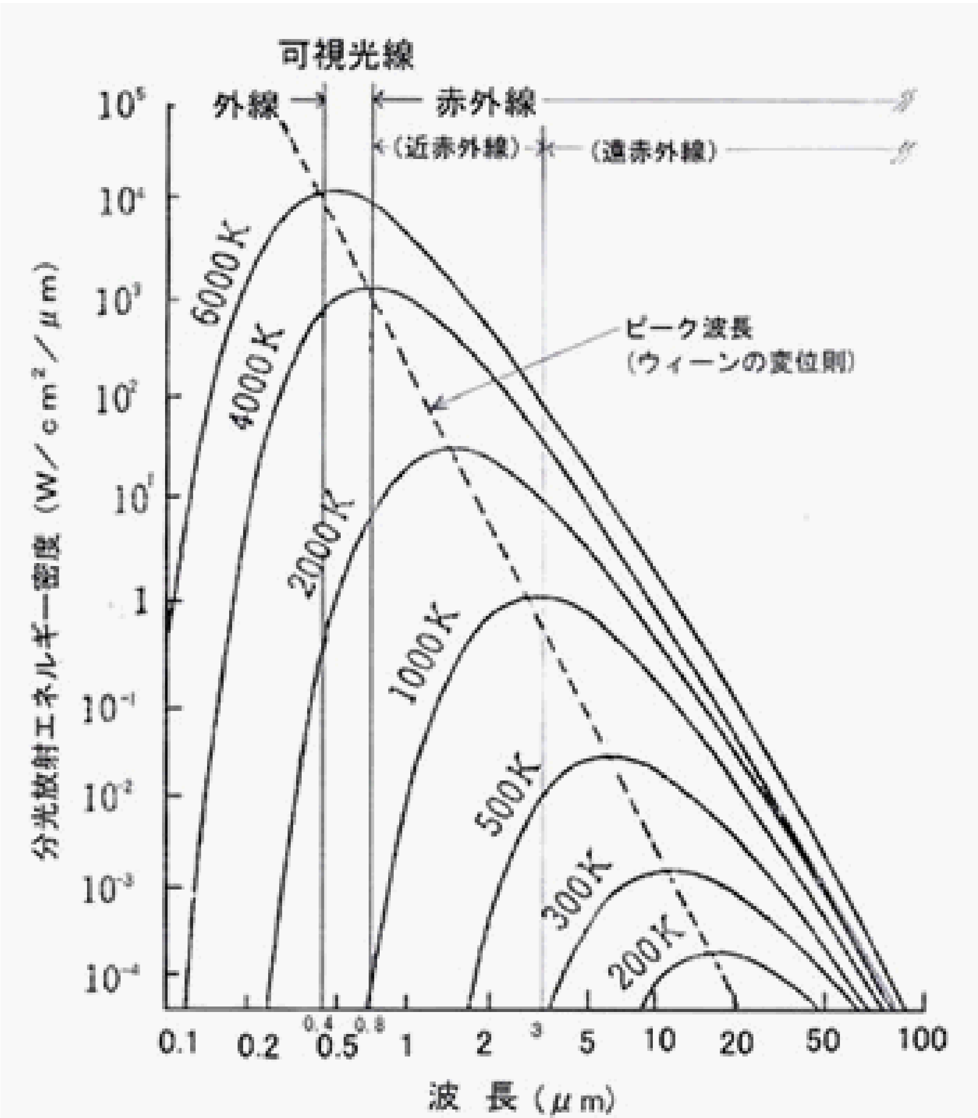
Snakes can “see” infrared light.

Pit organs (membrane) detect thermal infrared radiation.
(~1 m distant)



Revolution for effective hunting

Temperature and irradiation wavelegnth



Plank’s law

$$I(\nu,T) = \frac{2h\nu^3}{c^2}\frac{1}{e^{h\nu/kT}-1}$$

Human emits far infrared light with a peak wavelength at 10 μm.

Heat can be detected with infrared light.

Contents

Learn from eye in nature

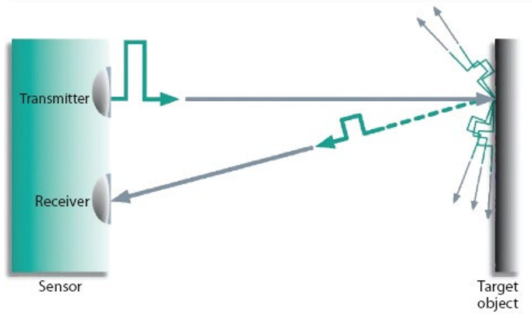
- Natural selection leaves living thins which are well adapted to their environments.
  - Bees use UV to effectively obtain pollen.
  - Flowers evolved to be able to be seen in UV to maximize pollination opportunity.
  - Snakes evolved to be able to detect infrared light to effectively hunt.
- No need to be constrained in visible world
  - We do not need to limit usable wavelength to be visible when we construct intelligent mechanics like robots.
  - Use of other wavelength then visible may lead to ***good object recognition***

Current sensing technology in machine vision

Single eyed or stereo camera

Visible･Near-infrared
Image recognition using camera. Stereo camera provide ***three dimensional recognition*** using parallax.

Laser range finder



Time of flight using pulsed infrared laser for recognition of ***distance data*** of outer world.

Milli rader

Rader using millimeter RF wave for obtaining ***distance data***
~100m-distance
(Automotive applications)

Infrared, and milli-wave for distance Recognition
Visible for image recognition

What will happen if we use other Wavelength? What kind of merit?

Advantages of using other wavelengths

(Ex) Automatic automotives

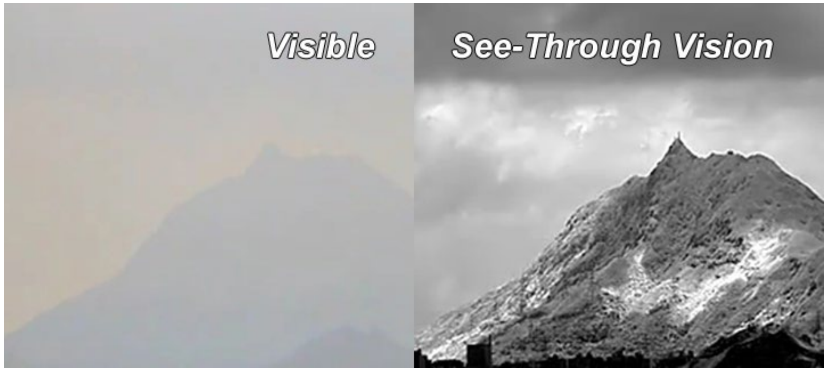
Visible camera for environment recognition

Issue

Rain and snow disturb the recognition.

Human or mannequins?

Near IR camera http://www.kaya-optics.com/



Less suffering from scattering

Recognition is easy.

Use of Infrared is advantageous for object recognition.

Advantages of IR in automotive use

対象	想定される(誤)認識シーン	検知方法	波長			
			可視	近赤	中赤	遠赤
縁石 	路面の両サイドにある縁石 夜間、日中(日射が強い時)	アスファルトとの違い	△	○ 1.7μm	○ 3.4μm	△
ガードレール 	背景との同化					
	木々	木々との違い	△			○
	歩道のアスファルト	アスファルトとの違い	△	×	○ 3.4μm	○
	白っぽい建物(コンクリート)	コンクリートとの違い	△	○ 1.7μm	×	○
	自動車	自動車(塗料の違い)	△	○ 1.7/2.2μm	○ 4.5μm	○
タイヤ片 	アスファルトとの同化	アスファルトとの違い	△	×	×	○
カラーコーン ポールコーン 	赤、緑で構成されているため 誤認識小	アスファルト、 コンクリートとの違い (夜間など)	△	○ 1.7μm 2.2μm	○ 3.4μm 4.5μm	×
車止め 	アスファルト、 コンクリートとの同化	アスファルトとの違い (再生プラ)	△	○ 1.2μm	×	△
		コンクリートとの違い (ステンレス)	△	×	×	○

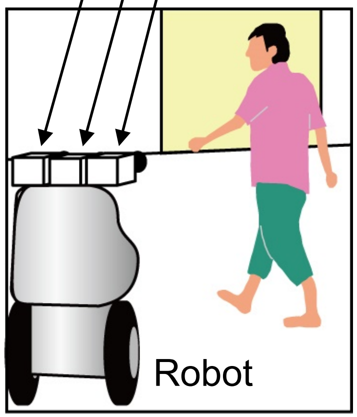
デンソー調査による

Issue to be solved in Vis/IR integrated camera

Conventional

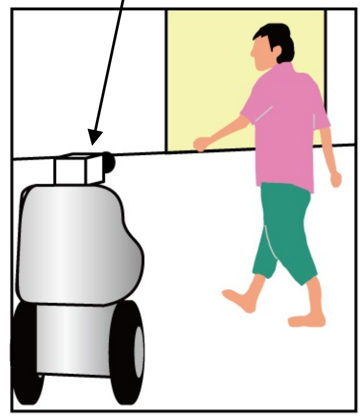
Multiple cameras were separately installed

Near IR camera
Far IR camera
Visible camera



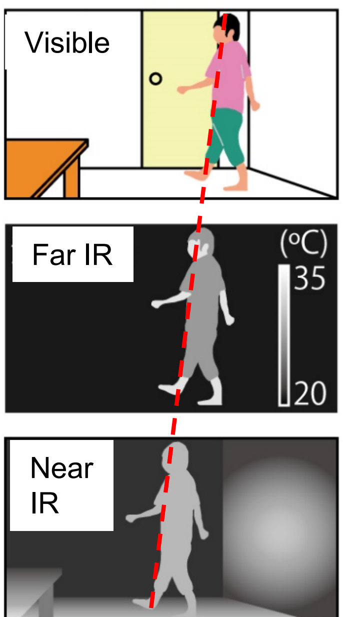
Proposed

Proposed multi wavelength camera



Co-axial & Same magnification

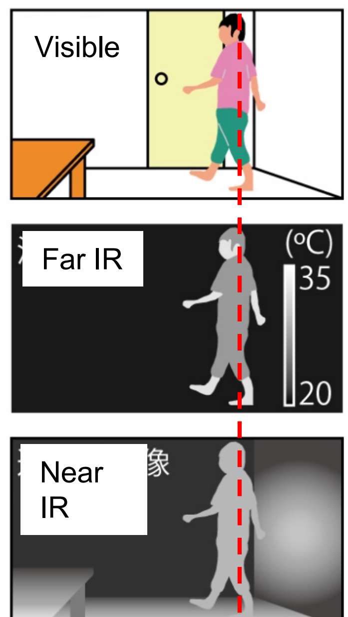
Issue



Lag happened

Pixel-to-Pixel correspondence between cameras

Magnification difference between cameras



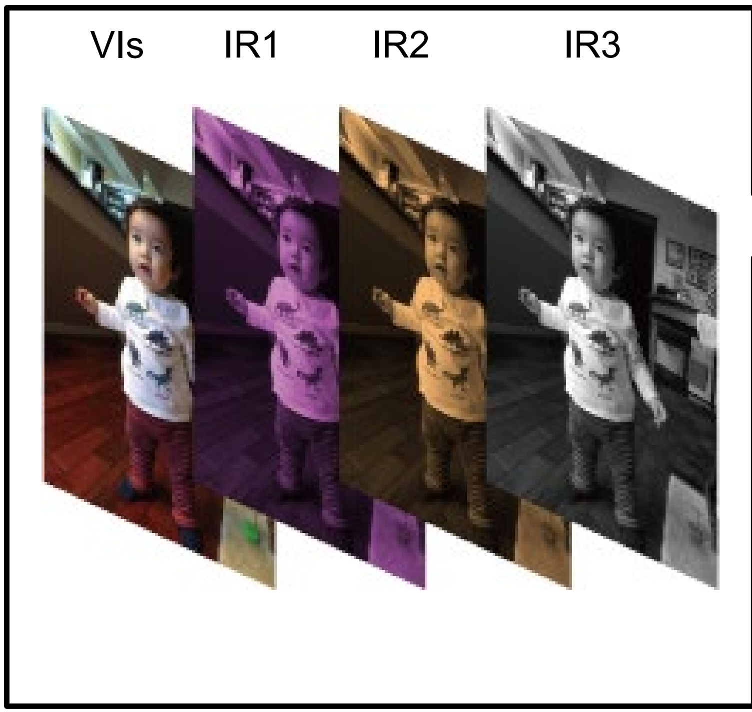
No lag

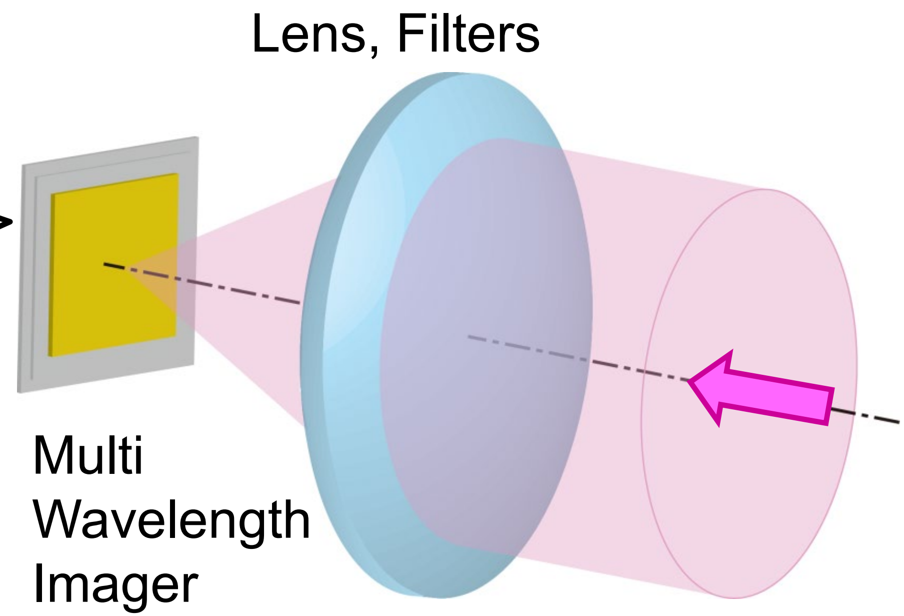
Pixel-to-pixel correspondence is assured in a hardware level.

Fast and precise recognition.

Robot vision hardware

The same optical axis, magnification, multi-wavelength camera
IR image can be directly **overlapped on a visible camera images.**



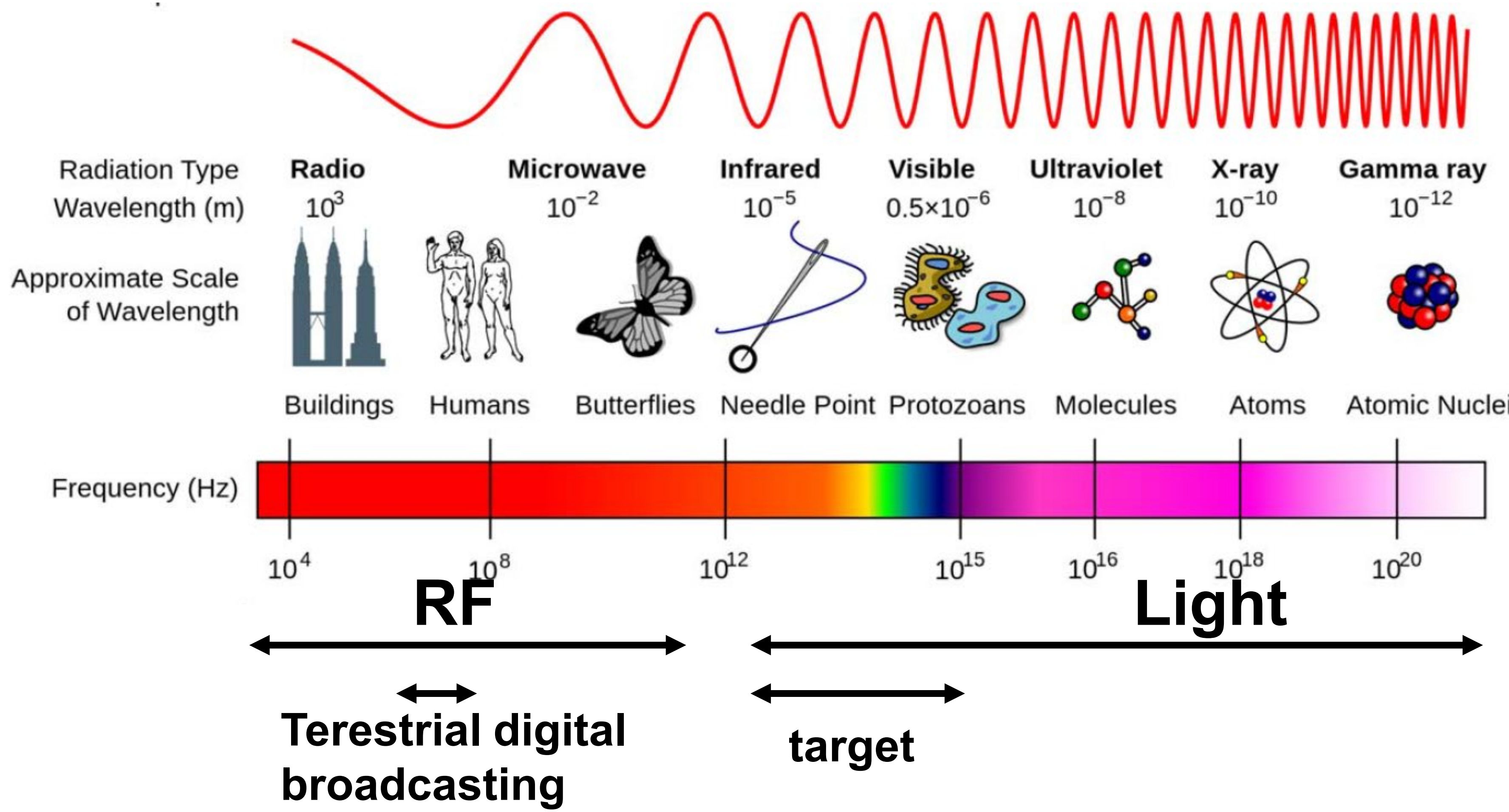


Vis~IR multi-wavelength

- **Precise and High** Speed Object Recognition
- **Material can also be analyzed** by the spectrum data

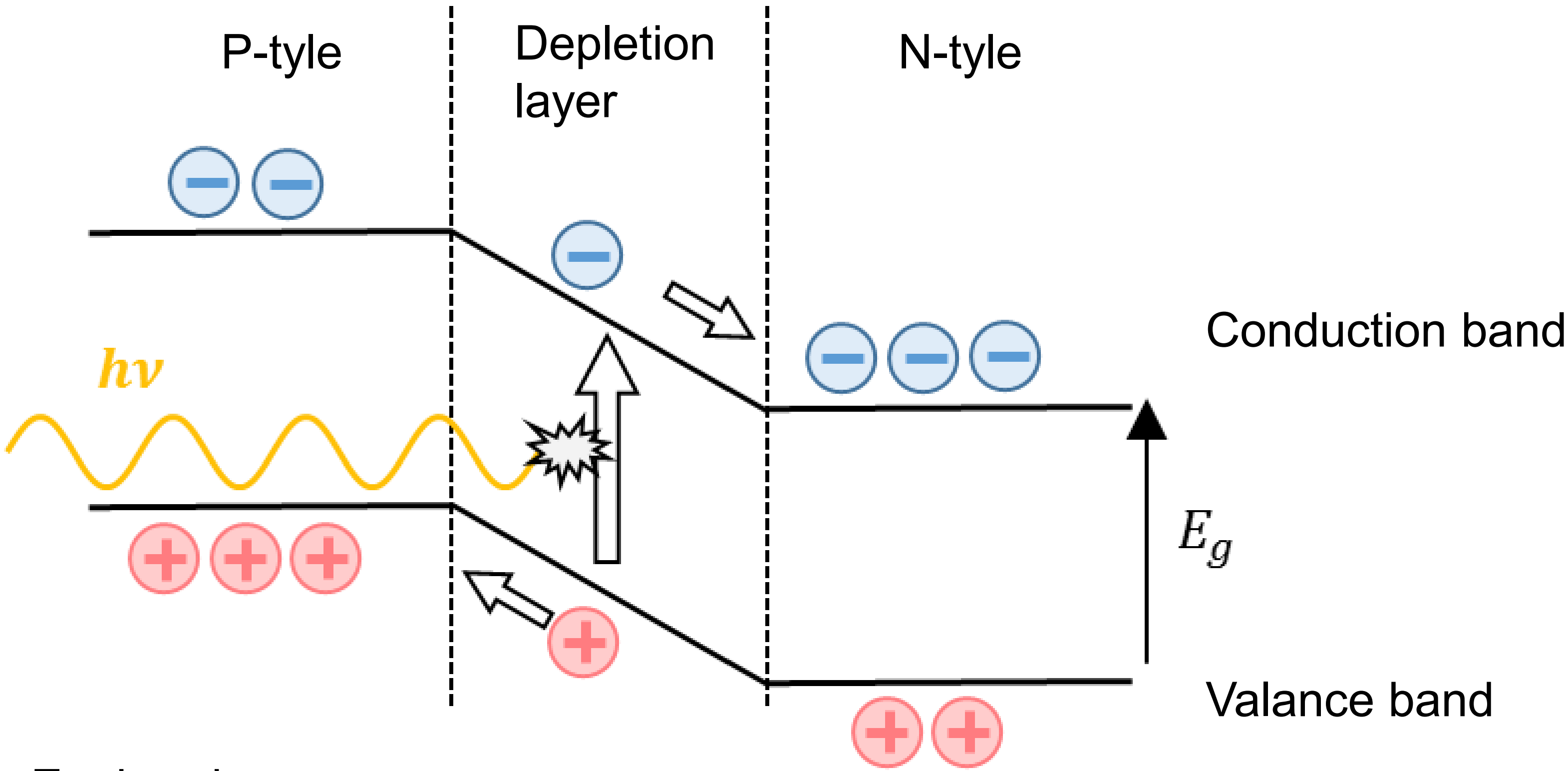
Contents

Wavelength & frequency



- Vis：500~700 nm, Near IR: 1~2.5μm, Mid IR: 2.5~4μm Far IR: 4~10μm
- (cf) Terestrial digital broadcasting 500MHz (λ~60cm)

Light detection using semiconductor



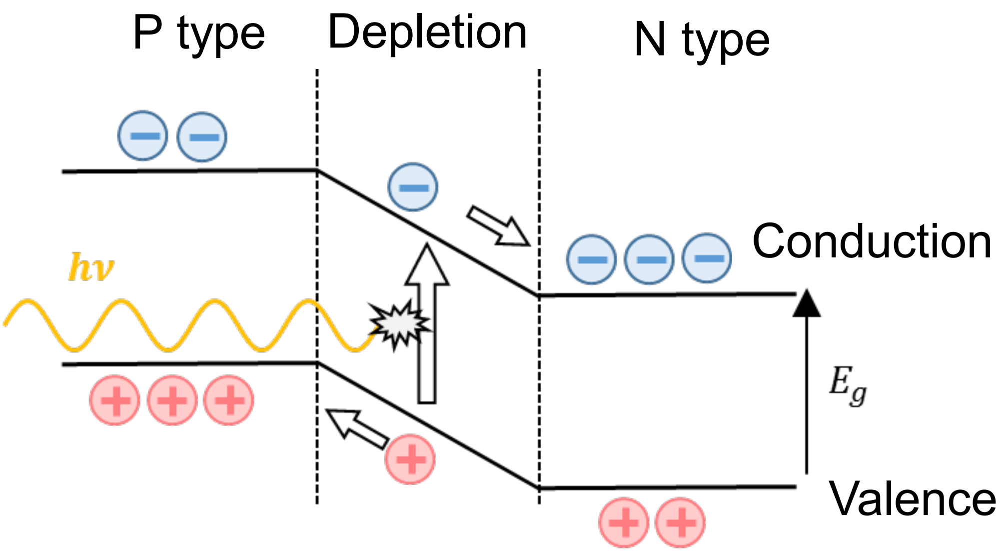
Eg: bandgap energy
hv: quantum energy of photon

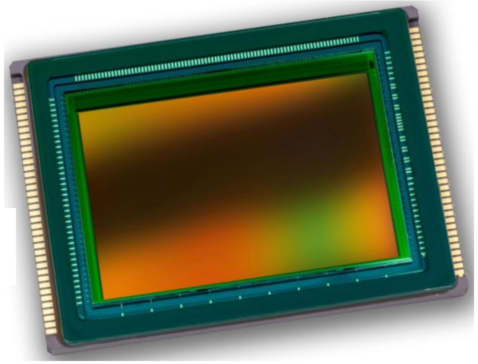
http://optipedia.info/

For photodetection, $hv > E_g$

Silicon is versatile, but not good at IR detection

Material	bandgap (eV)	Max λ (μm)
Si	**1.11**	**1.12**
InGaAs	0.5~0.7	1.8~2.5
InSb	0.17	7.3





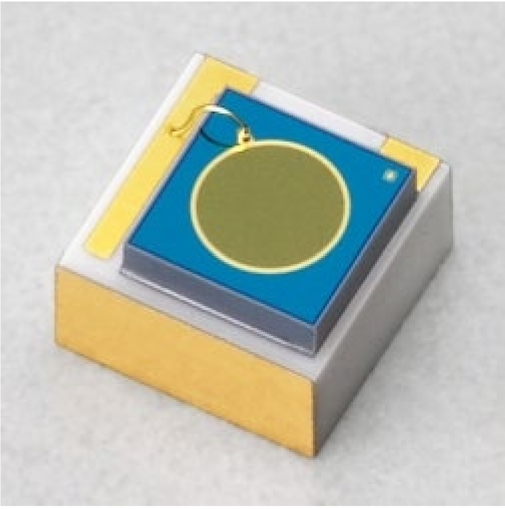
Visible detectors are made of Si.

http://www.cmosis.com

λ over 1.1 μm **cannot be detected** with solely Si.

- We want to construct Si based IR photodetector
  - Low cost and **high affinity with CMOS and MEMS**.
- How to detect IR with Si?

従来の赤外光ディテクタ(量子型)



InGaAsフォトダイオード
(インジウムガリウムヒ素)
0.9～1.7μm対応

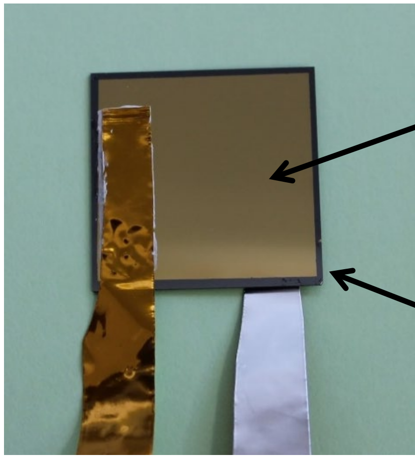
InAsSb光起電力素子
(インジウムヒ素アンチモン)
～5μm対応

InSb光起電力素子
(インジウムアンチモン)
3~5 μm対応

- 主に化合物半導体が使われている
- シリコンで波長選択的な赤外光ディテクタができれば、可視光ディテクタと混在可能
- 材料物性に頼らず、どのように実現?

Infrared Detection with Si

Metal film (gold) on Si



Metal species and
barrier height Φ_B (eV)

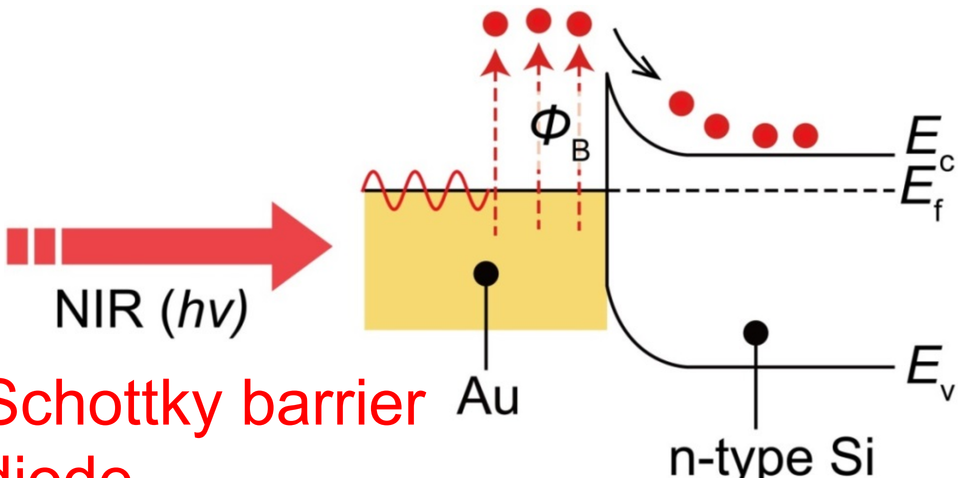
Metal	Φ_B (eV)
Chromium	0.45
Gold	**0.65**

E_g (1.1 eV) > Φ_B

Detectable limit increased.

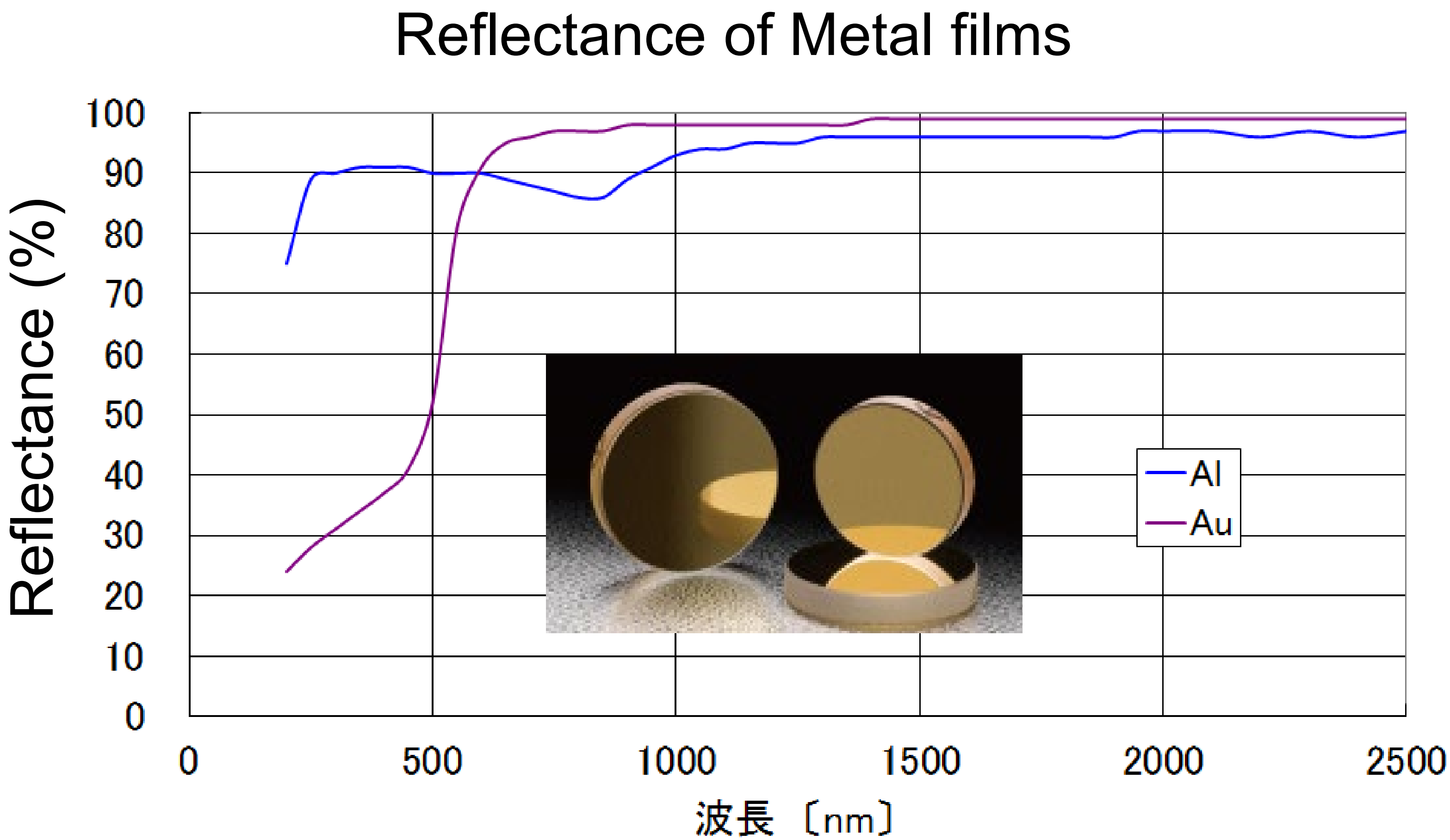
Infrared detection **possible with a Si device**.

Schottky barrier Φ_B



Schottky barrier diode

However, the gold is reflective media.



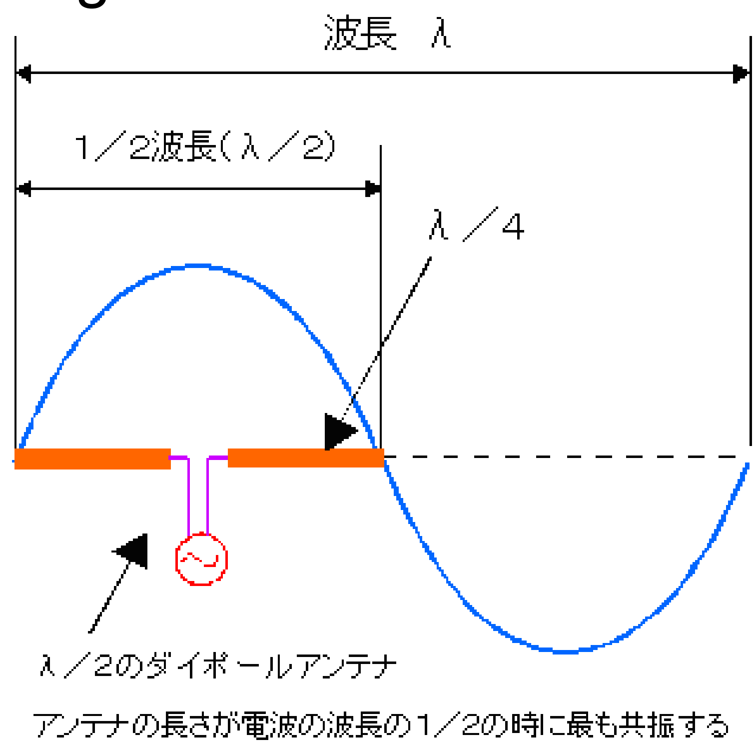
http://www.sigma-koki.com/

Although photo energy should be absorbed, the gold surface present high reflectivity. ⇨ **Issue to be solved.**

Light absorption using antenna

- Antenna for RF waves
  - Antenna length ~ RF wavelength

https://ja.wikipedia.org/wiki/八木・宇田アンテナ



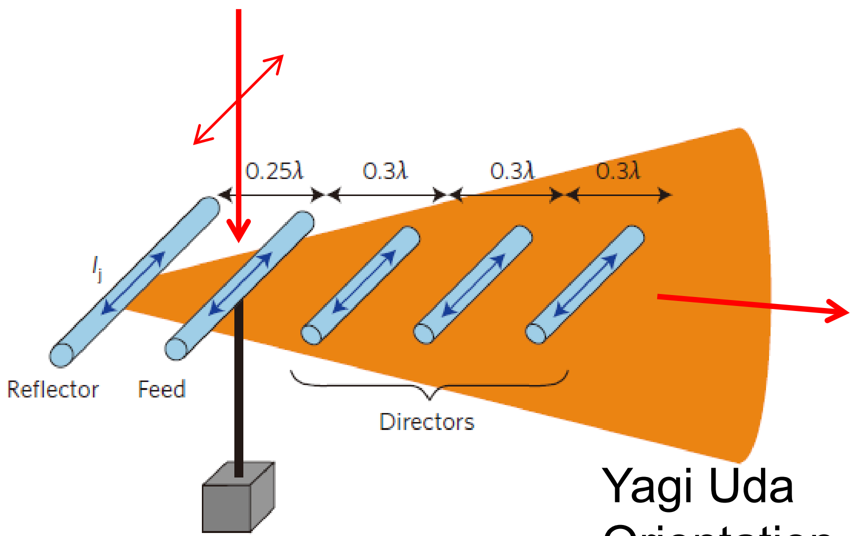
http://www.circuitdesign.jp/jp/technical/DesignGuide/guide4.asp

Antenna determines the wavelength of the absorbed light

Nanometric antenna of Vis/IR absorption

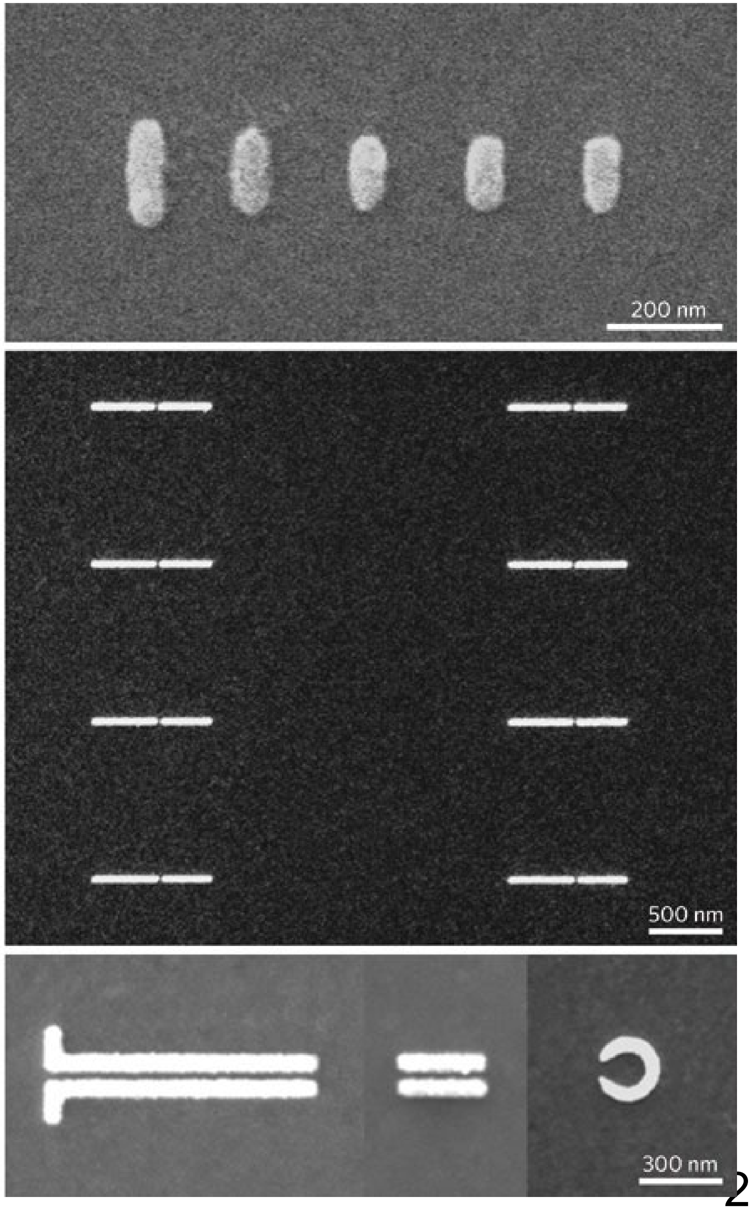
- Antenna comparable to wavelength.
- They work as antenna for light.

アンテナを電磁波で励振

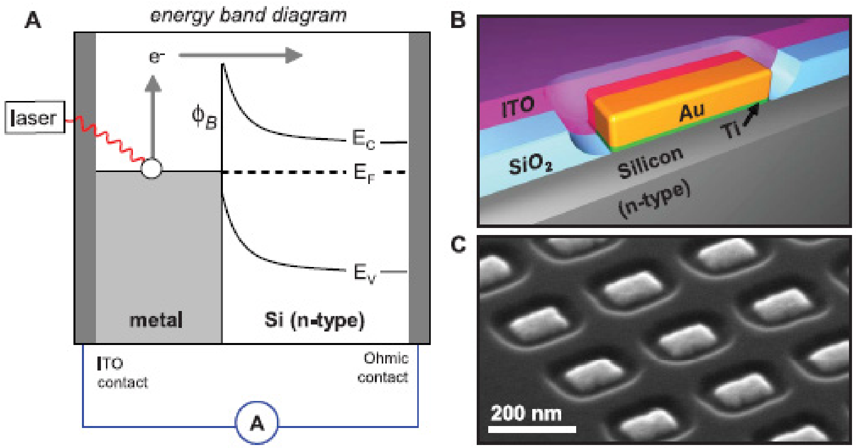


Yagi Uda Orientation

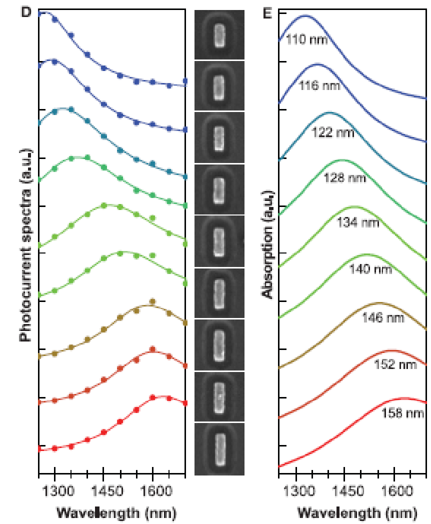
Lukas Novotny & Niek van Hulst, Nature Photonics 5, 83–90 (2011)



Absorption enhancement by nano antennas



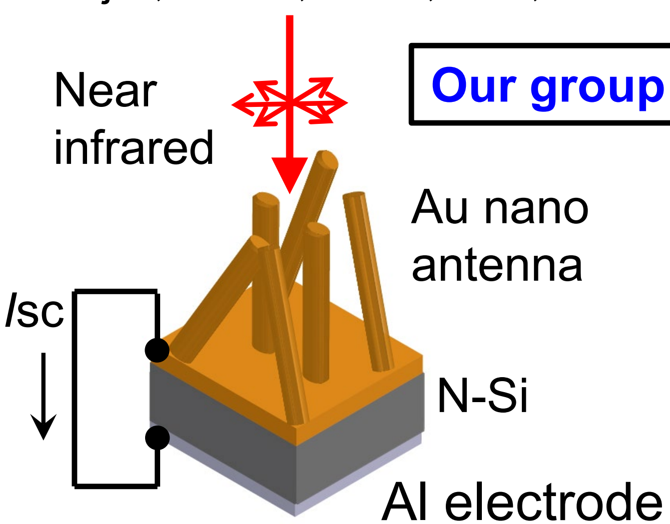
Knight, *et al.*, *Science*, 2011.

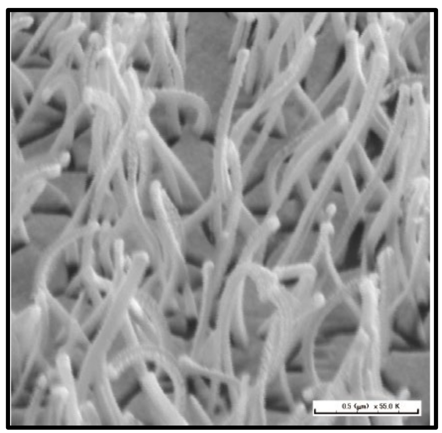


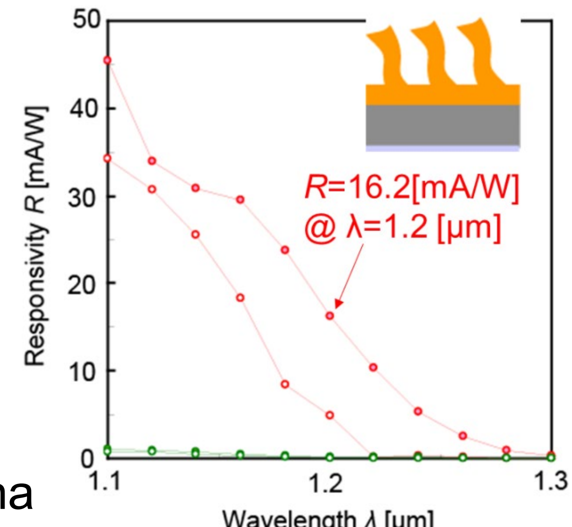
Plasmonic antenna
High precision nanofab is required.

Y. Ajiki, T. Kan, *et al.*, *APL*, 2016.

Our group





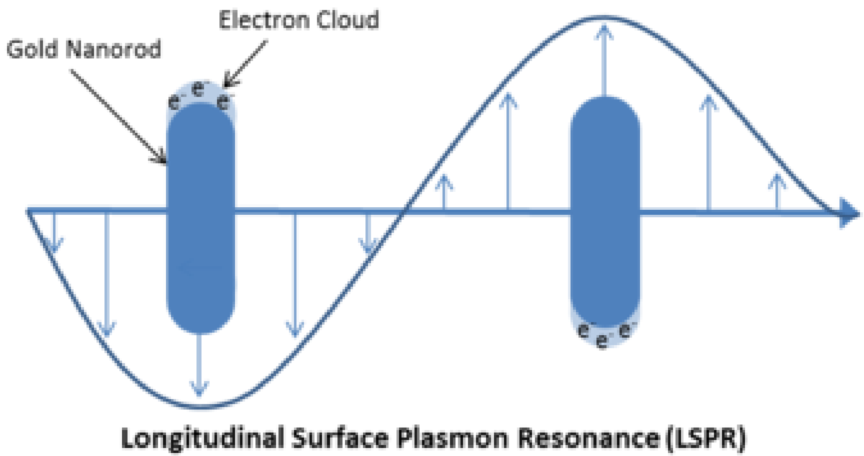


Self assembled antenna enhanced infrared absorption

However, random and difficult for mass production

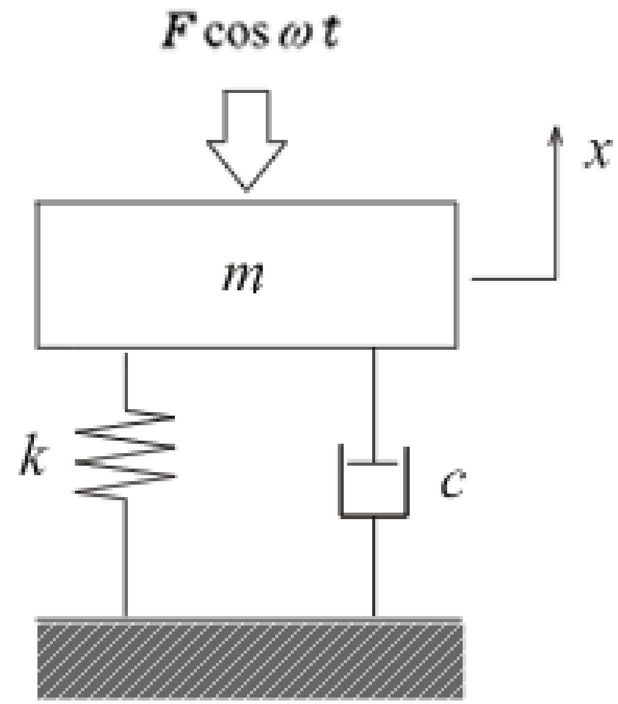
Absorption mechanism

- Inside the antenna, free electrons of metal antenna oscillates (Surface plasmon resonance)



http://nanohybrids.net/pages/plasmonics

Analogy

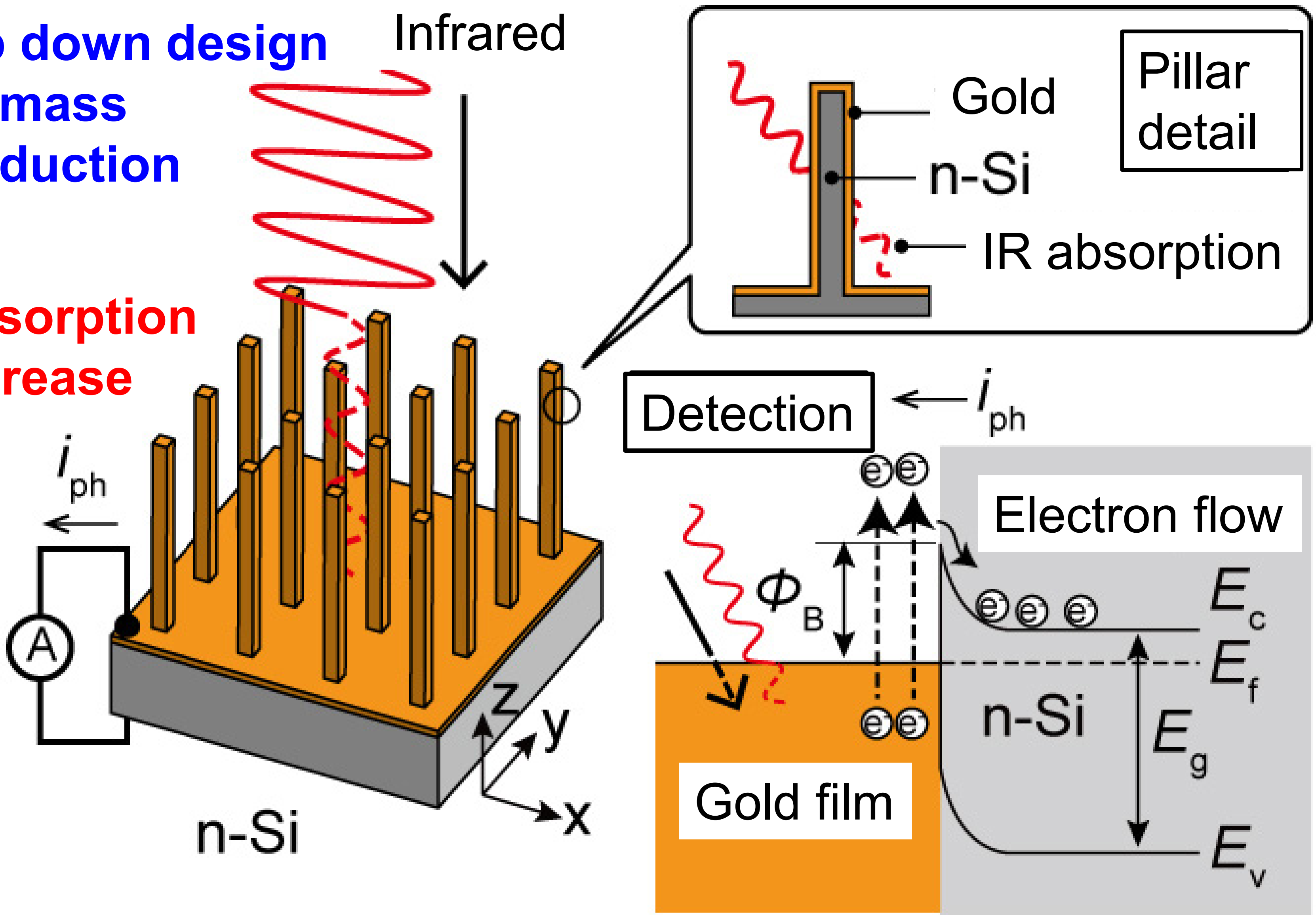


It can be regarded as spring-damper-mass system.
The antenna structure determines the resonance freq.

Si MEMS Near-Infrared Photodetector

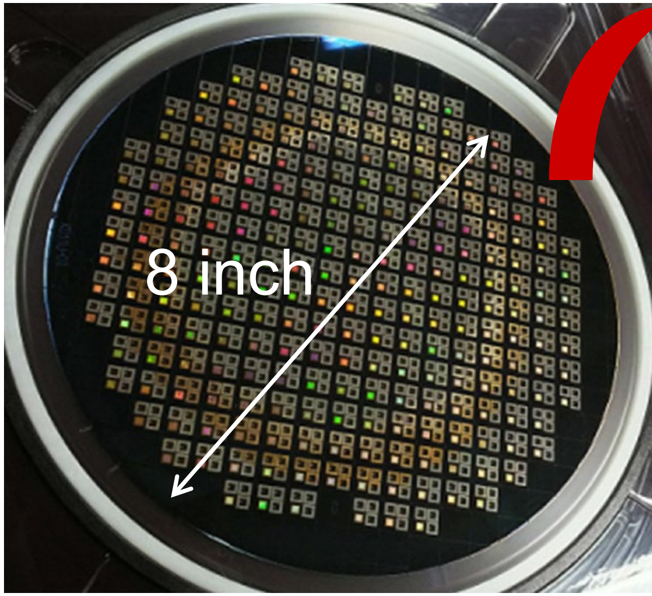
Top down design for mass production

① Absorption increase

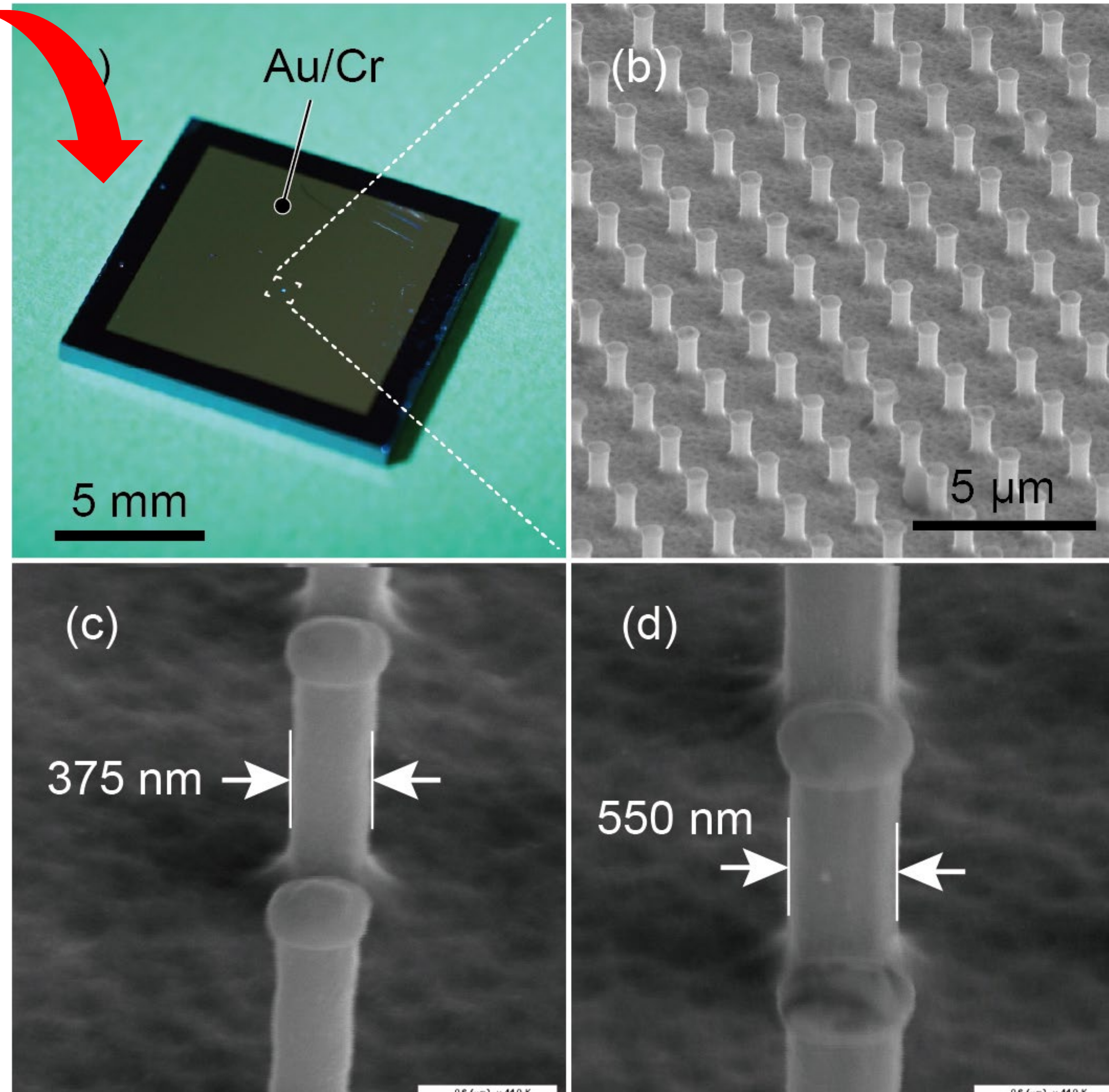


② Photodetection as electric signal by Schottky barrier

Uniform antenna fabrication using MEMS line



8 inch Si fabrication line
@Tsukuba, AIST, Japan

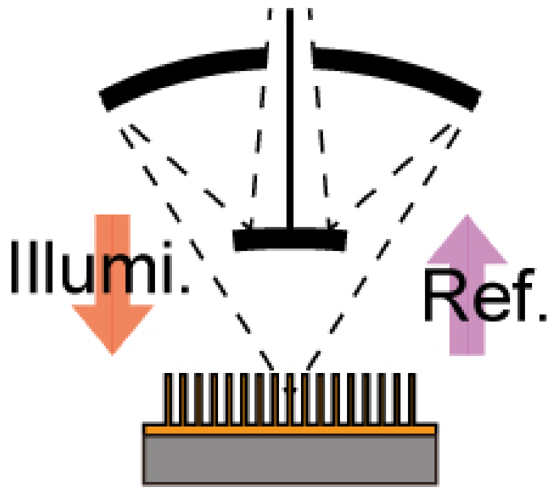


Pillar height : 1 μm
Pillar pitch : 2 μm
Au film : 20～30nm (side)
Diameter: 375 and 550 nm

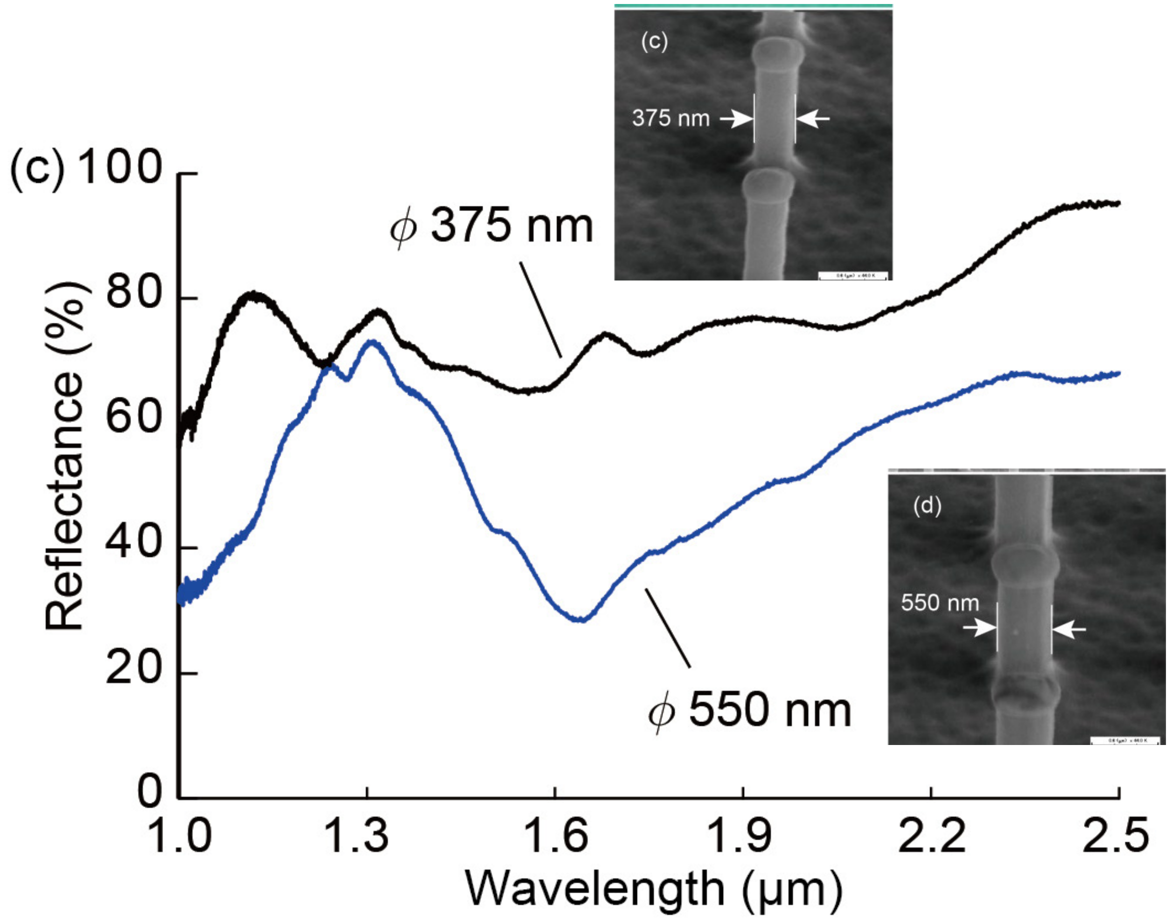
Good uniformity

Infrared absorption characteristics

FTIR Spectroscope Microscopy

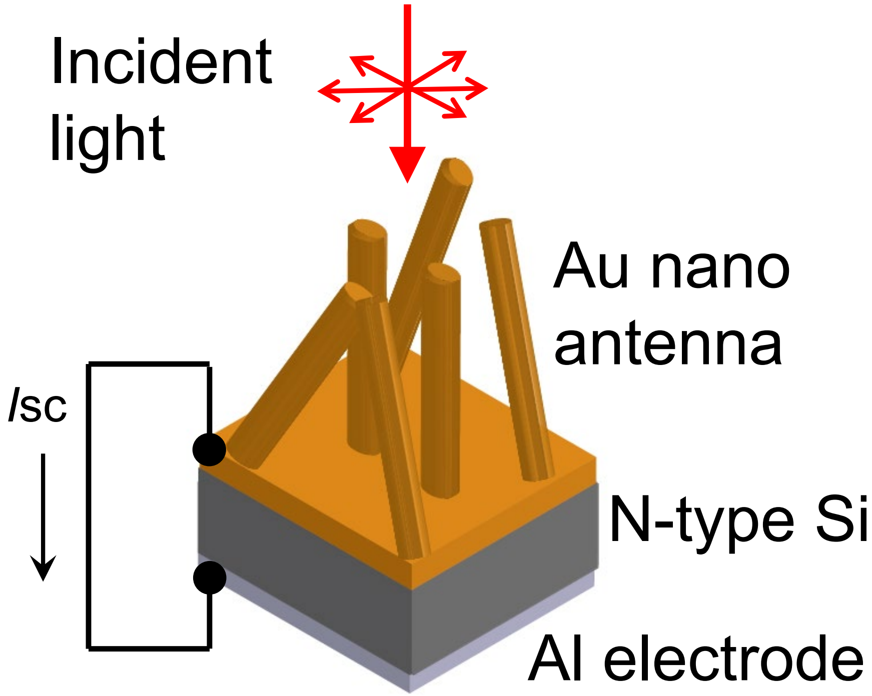


Nano-pillar

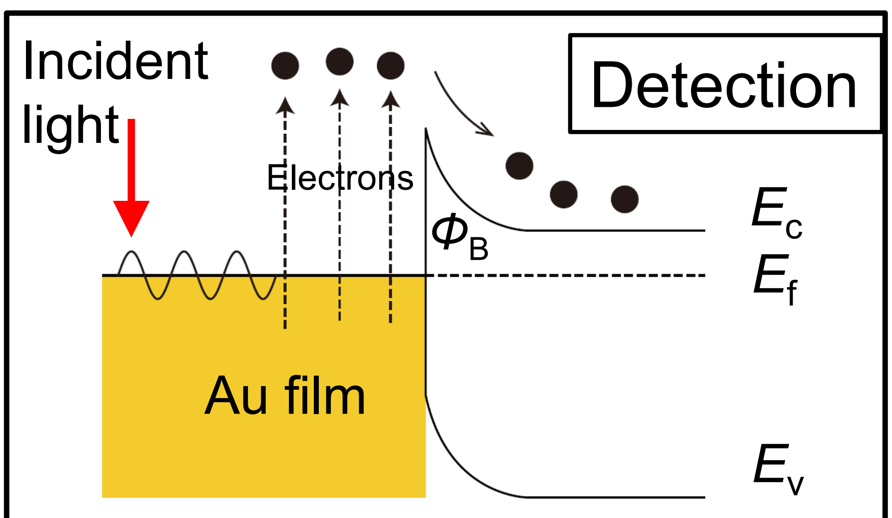


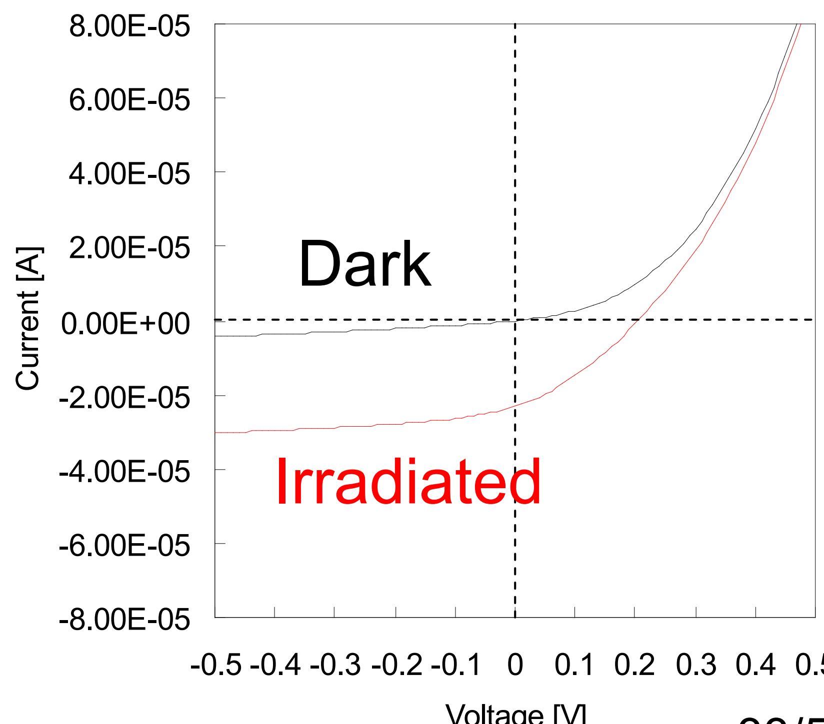
Broad absorption characteristics were observed for **λ 1.0 ~2.5 μm**. **50% infrared energy** was thought to be **absorbed** by the antennas.

Current detection via SPR



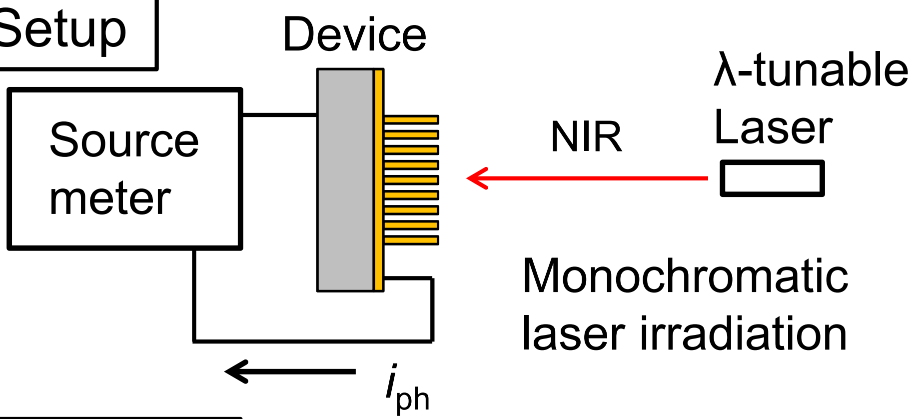
- Si can not solely detect infrared light（λ > 1.1μm）.
- Nano antenna light absorption
- Schottky barrier ~0.7eV for current detection



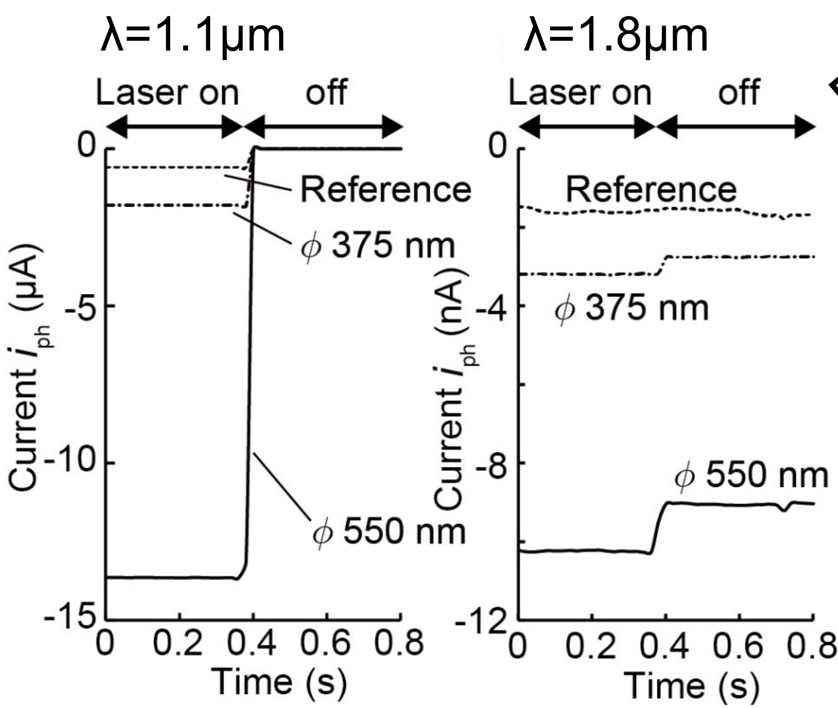


Photodetection characteristics

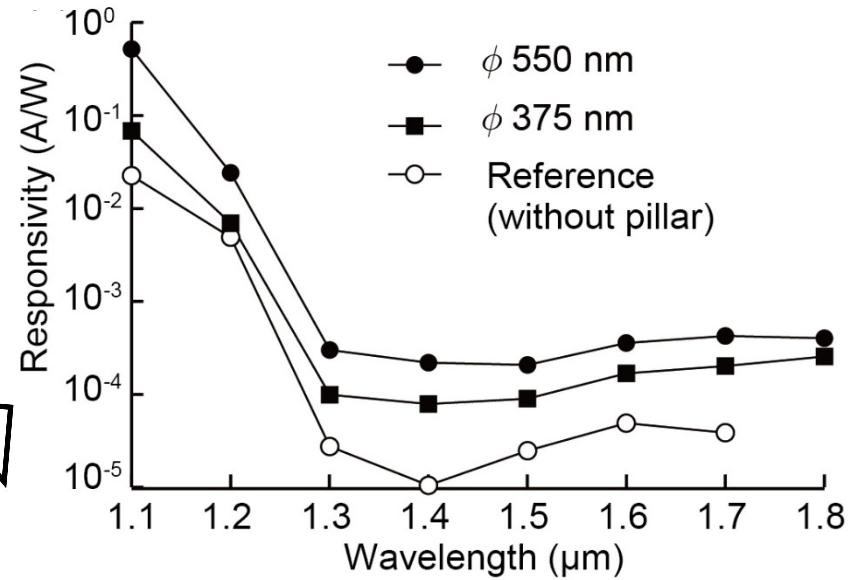
Setup



Raw data

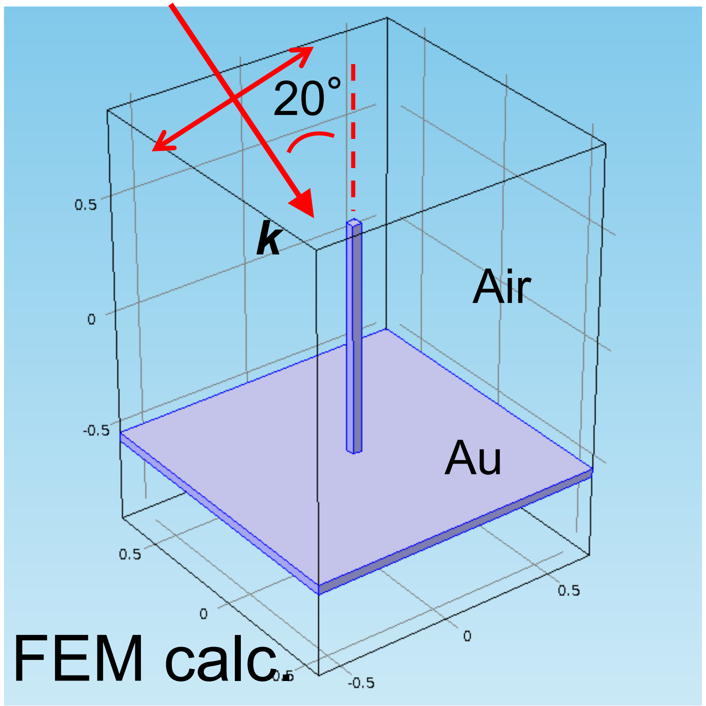


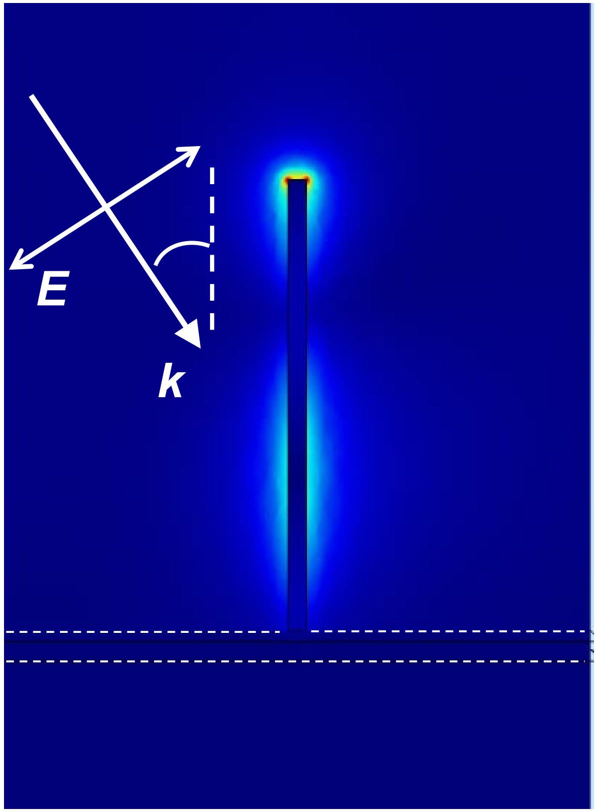
Sensitivity



- **10-times increase of sensitivity** due to nano-pillar antenna
  - Control device could not detect λ = 1.8 μm due to low S/N
- Detectable wavelength ~**2.1 μm** (so far).

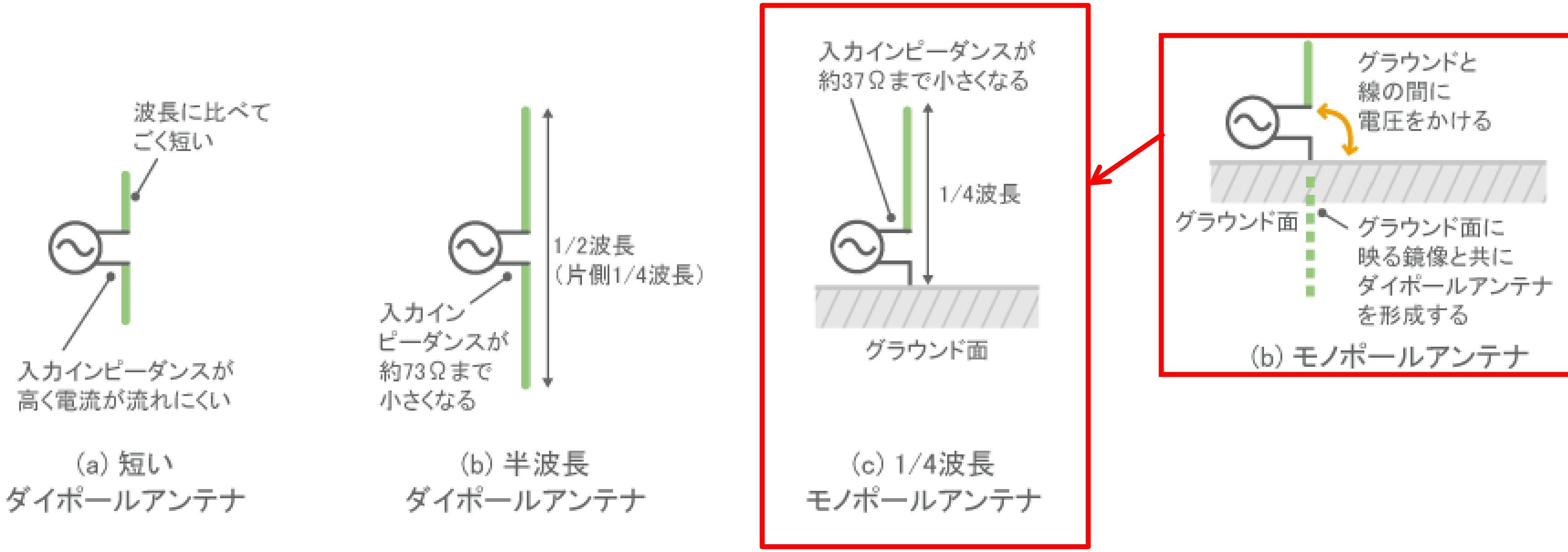
Detailed absorption mechanism



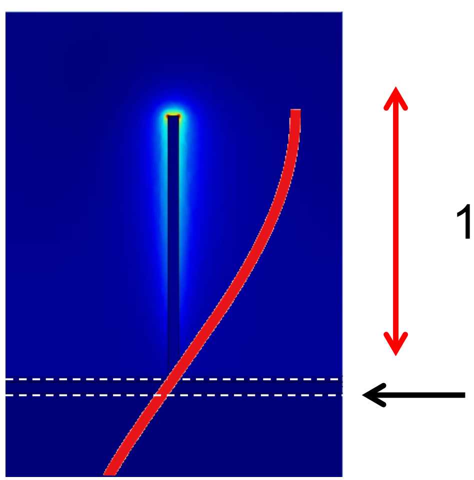


Electric field norm.
Charge accumulation at the antenna end.

Monopole antenna behavior



http://www.murata.com/ja-jp/products/emc/emifil/knowhow/basic/chapter04-p2



1/4 wavelength

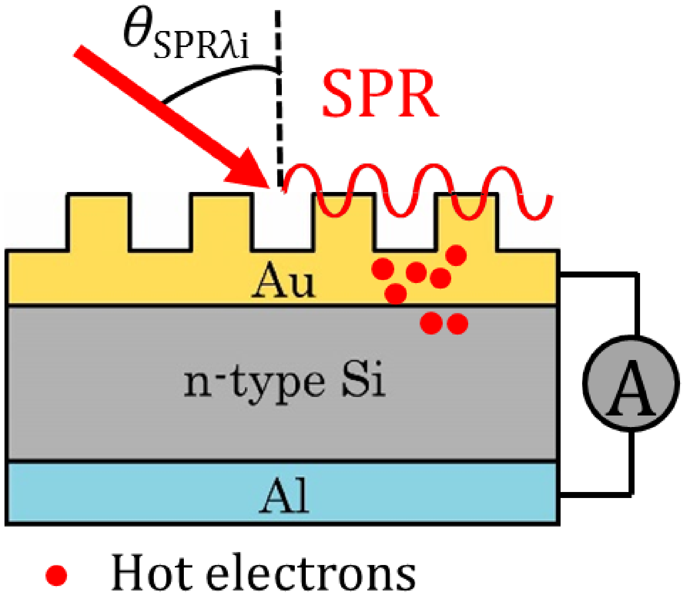
The Au film works as a mirror surface.

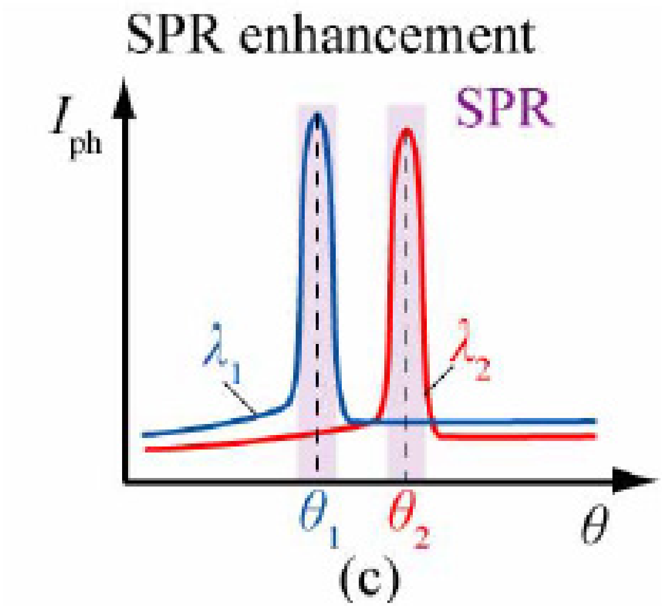
Contents

1. Learning from living things
2. Hint from living things
   Functions of “eye” in intelligent machines
3. Detection of infrared light
4. **Spectroscopic detection of light**
5. Polarization specific detection
6. Summary

Surface plasmon resonance on grating

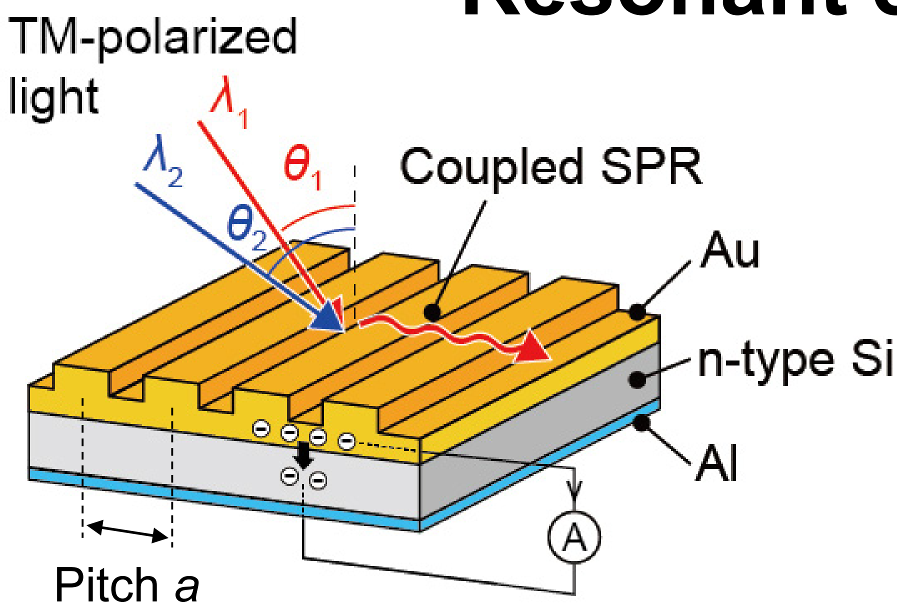
- Electron resonance at metal & dielectric interface
- Excitation by irradiation on the metal grating
- Sharp resonant nature (wavelength & angle)





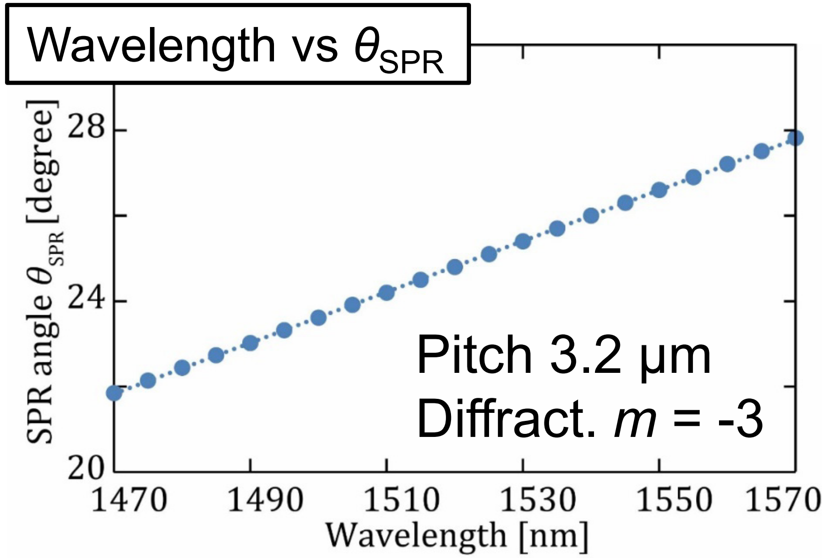
The resonant condition can be monitored electrically.

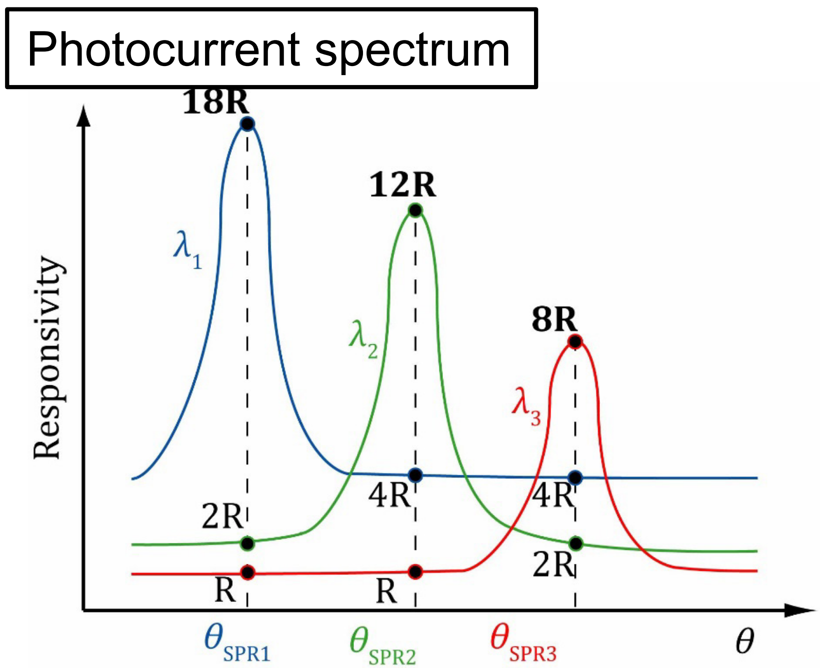
Resonant condition



$$\frac{2\pi}{\lambda_0}\sin\theta + \frac{2m\pi}{a} = \frac{\omega}{c}\sqrt{\frac{\varepsilon_{\mathrm{m}}+\varepsilon_{\mathrm{Au}}}{\varepsilon_{\mathrm{m}}\varepsilon_{\mathrm{Au}}}}$$

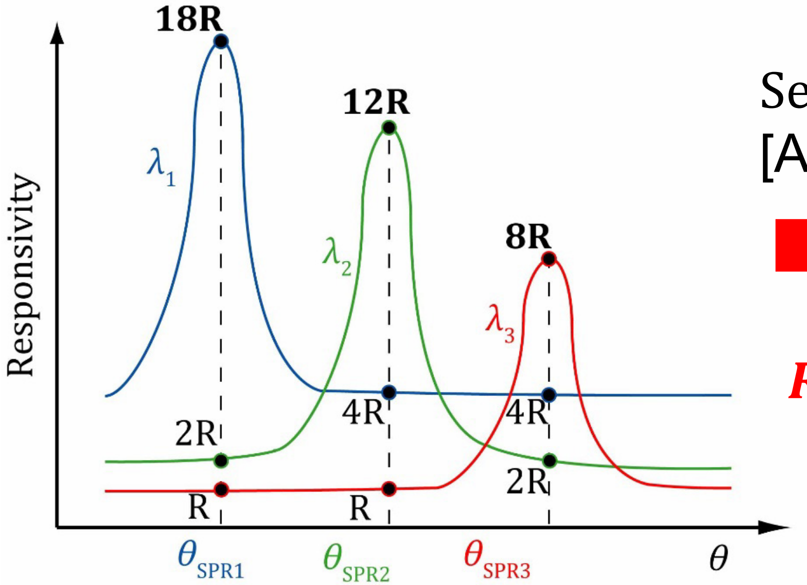
- Current peaks appeared for specific angle θ_{SPR}
- θ_{SPR} different depending on the wavelength





Spectrum Calculation Algorithm

Photocurrent spectrum



Sensitivity [A/W]

$$R = \frac{I_{ph}}{P_{in}}$$

Sensitivity matrix of the device

Responsivity matrix **R**

$$\begin{bmatrix} \mathbf{18R} & 2R & R \\ 4R & \mathbf{12R} & R \\ 4R & 2R & \mathbf{8R} \end{bmatrix}$$

Current at diffent θ

Summation of responses for wavelengths

$$I_{\theta_{SPR1}} = R_{\lambda_1\theta_{SPR1}}P_{\lambda_1} + R_{\lambda_2\theta_{SPR1}}P_{\lambda_2} + R_{\lambda_3\theta_{SPR1}}P_{\lambda_3} + \cdots + R_{\lambda_n\theta_{SPR1}}P_{\lambda_n}$$

$$I_{\theta_{SPR2}} = R_{\lambda_1\theta_{SPR2}}P_{\lambda_1} + R_{\lambda_2\theta_{SPR2}}P_{\lambda_2} + R_{\lambda_3\theta_{SPR2}}P_{\lambda_3} + \cdots + R_{\lambda_n\theta_{SPR2}}P_{\lambda_n}$$

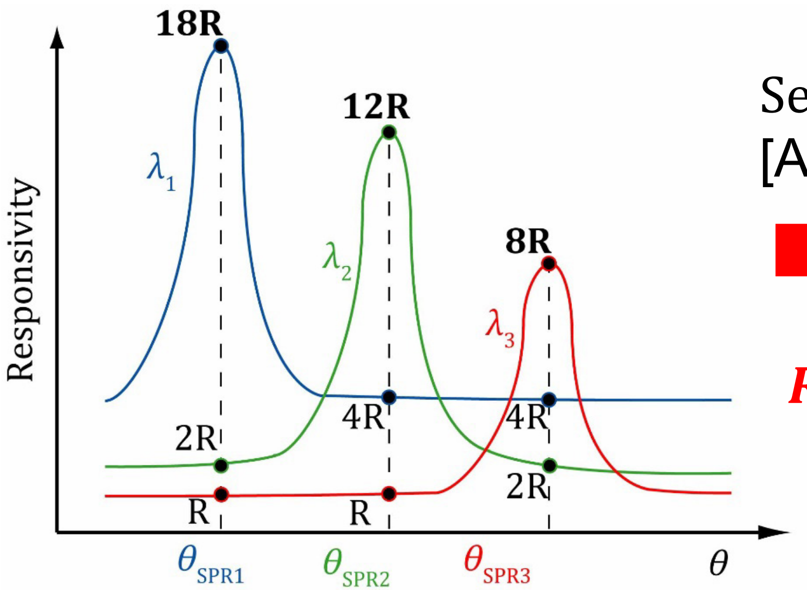
$$I_{\theta_{SPR3}} = R_{\lambda_1\theta_{SPR3}}P_{\lambda_1} + R_{\lambda_2\theta_{SPR3}}P_{\lambda_2} + R_{\lambda_3\theta_{SPR3}}P_{\lambda_3} + \cdots + R_{\lambda_n\theta_{SPR3}}P_{\lambda_n}$$

$$\vdots$$

$$I_{\theta_{SPRn}} = R_{\lambda_1\theta_{SPRn}}P_{\lambda_1} + R_{\lambda_2\theta_{SPRn}}P_{\lambda_2} + R_{\lambda_3\theta_{SPRn}}P_{\lambda_3} + \cdots + R_{\lambda_n\theta_{SPRn}}P_{\lambda_n}$$

Spectrum Calculation Algorithm

Photocurrent spectrum



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Sensitivity matrix of the device

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$$\begin{bmatrix} \mathbf{18R} & 2R & R \\ 4R & \mathbf{12R} & R \\ 4R & 2R & \mathbf{8R} \end{bmatrix}$$

Current | Responsivity matrix | input

$$\begin{bmatrix} I_{\theta_{SPR1}} \\ I_{\theta_{SPR2}} \\ I_{\theta_{SPR3}} \\ \vdots \\ I_{\theta_{SPRn}} \end{bmatrix} = \begin{bmatrix} R_{\lambda_1\theta_{SPR1}} & R_{\lambda_2\theta_{SPR1}} & R_{\lambda_3\theta_{SPR1}} & \cdots & R_{\lambda_n\theta_{SPR1}} \\ R_{\lambda_1\theta_{SPR2}} & R_{\lambda_2\theta_{SPR2}} & R_{\lambda_n\theta_{SPR1}} & \cdots & R_{\lambda_n\theta_{SPR2}} \\ R_{\lambda_1\theta_{SPR3}} & R_{\lambda_2\theta_{SPR3}} & R_{\lambda_3\theta_{SPR3}} & \cdots & R_{\lambda_n\theta_{SPR3}} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ R_{\lambda_1\theta_{SPRn}} & R_{\lambda_2\theta_{SPRn}} & R_{\lambda_3\theta_{SPRn}} & \cdots & R_{\lambda_n\theta_{SPRn}} \end{bmatrix} \begin{bmatrix} P_{\lambda_1} \\ P_{\lambda_2} \\ P_{\lambda_3} \\ \vdots \\ P_{\lambda_n} \end{bmatrix}$$

Known ***Known*** ***Unknown***

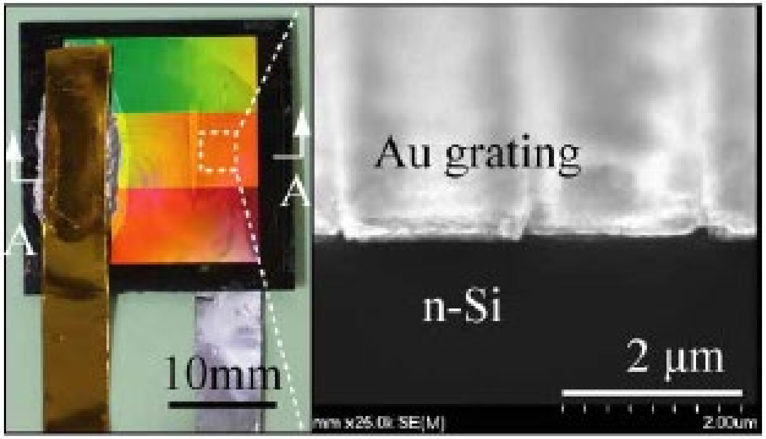
$$\boldsymbol{I = RP}$$

$$\Downarrow$$

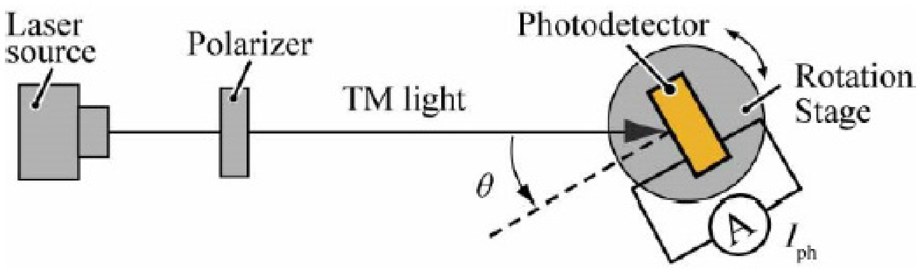
$$\boldsymbol{P = R^{-1} I}$$

Spectrum derivable

Check of hypothesis

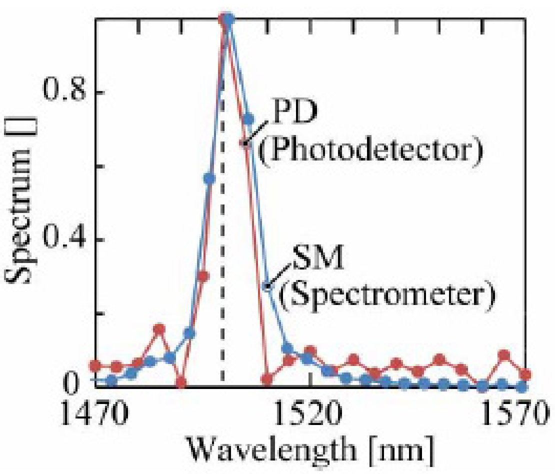


W.J. Chen, T. Kan, et al., *Optics Express*, 2016

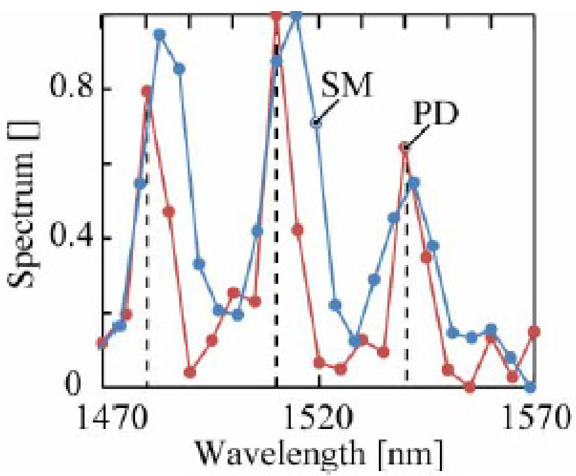


Rotation by a automatic stage

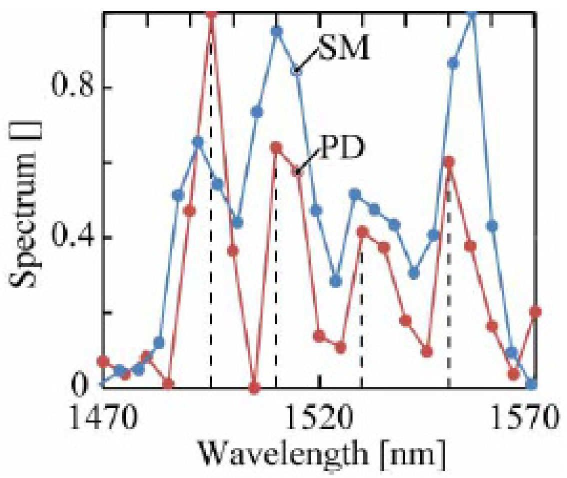
Input wavelength：1500 nm



1480, 1510, 1540 nm



1490, 1510, 1530, 1550 nm



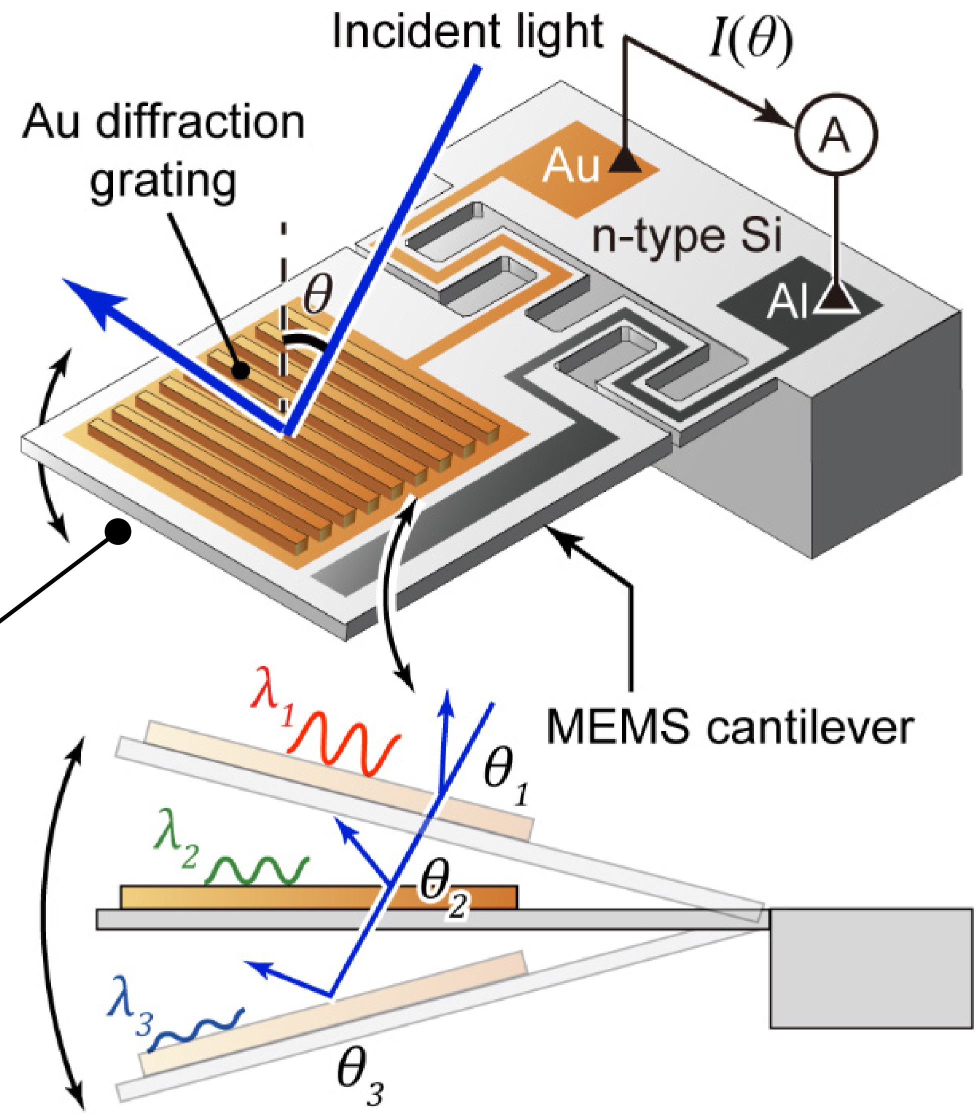
Comparable wavelength resolution with a commercial device

……*However, the whole system is still large.*

Toward a miniaturized spectrometer

One chip spectrometer with SPR modulation functionality

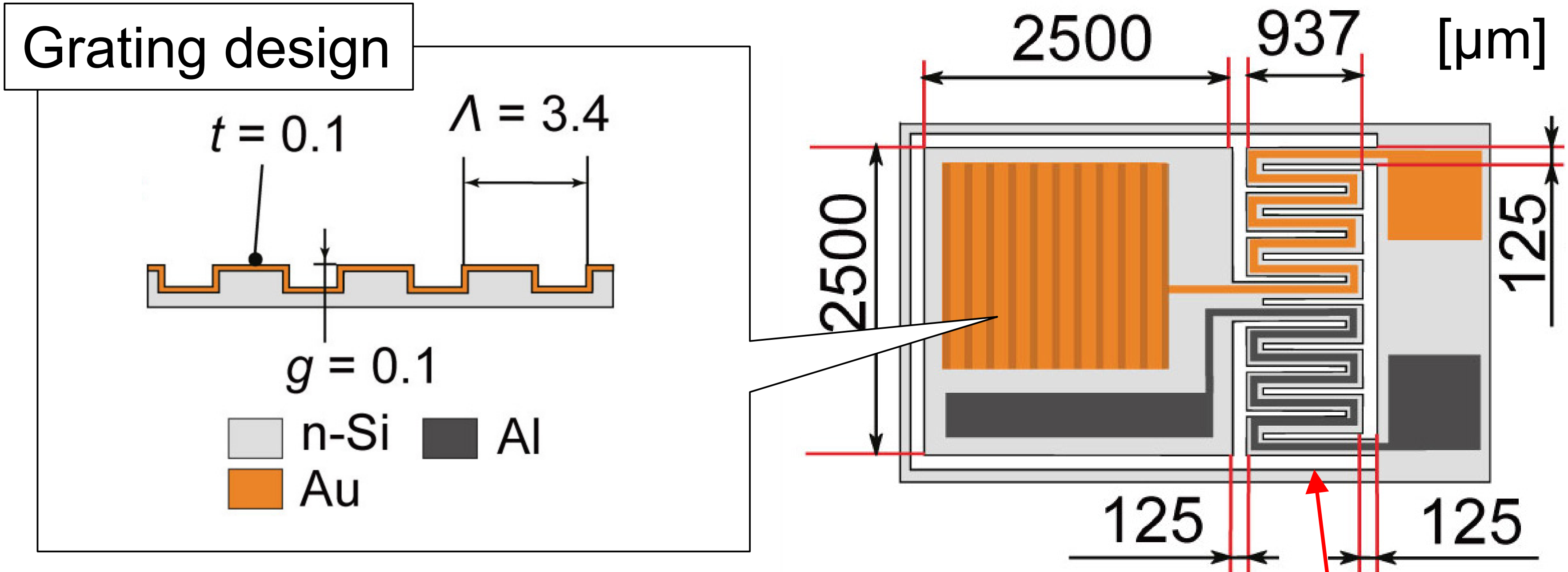
Ph.D student
Masaaki Oshita

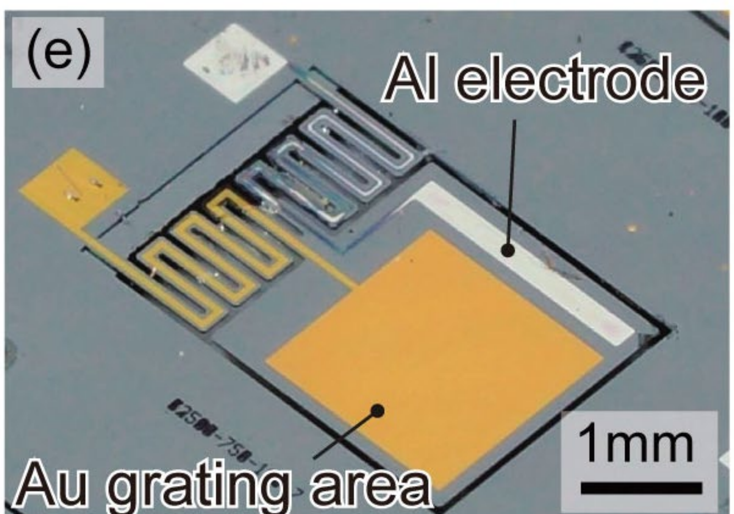


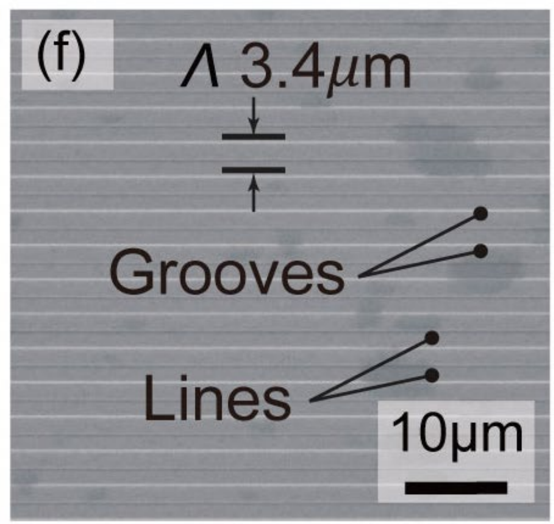
Grating on the MEMS tilting cantilever

Mechanical vibration **modulates the angle of incidence**

Photographs of MEMS device

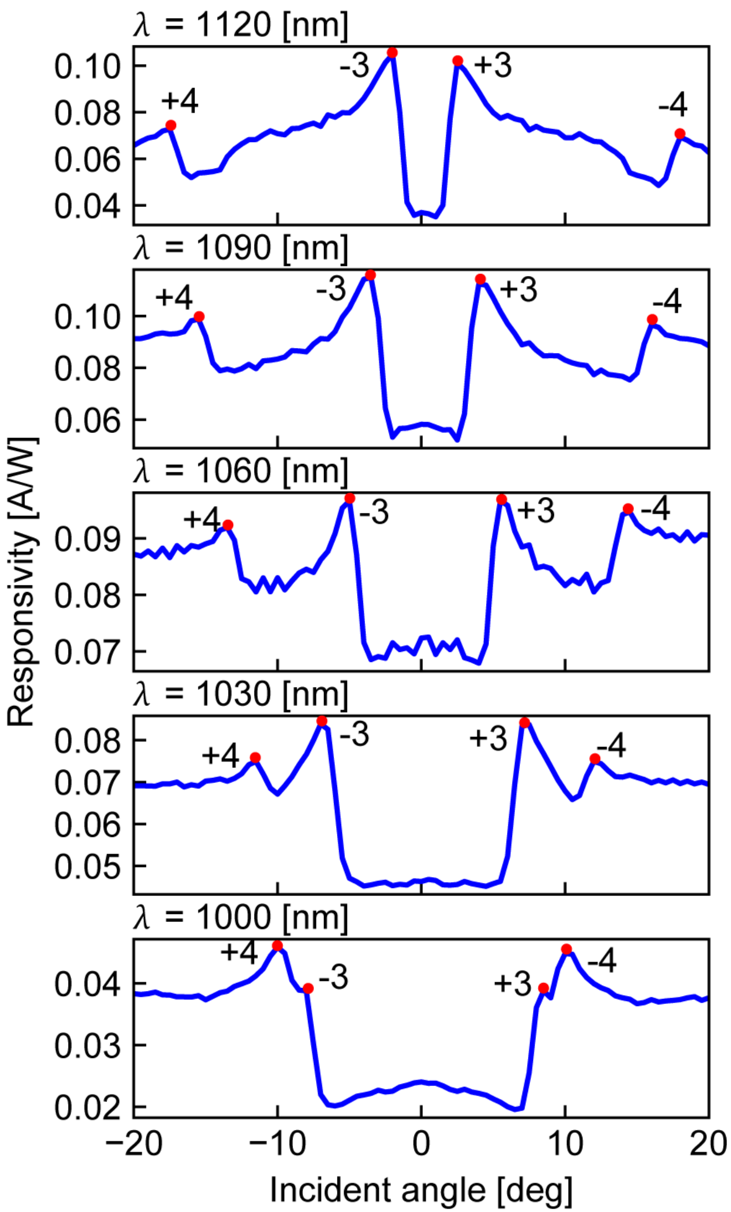


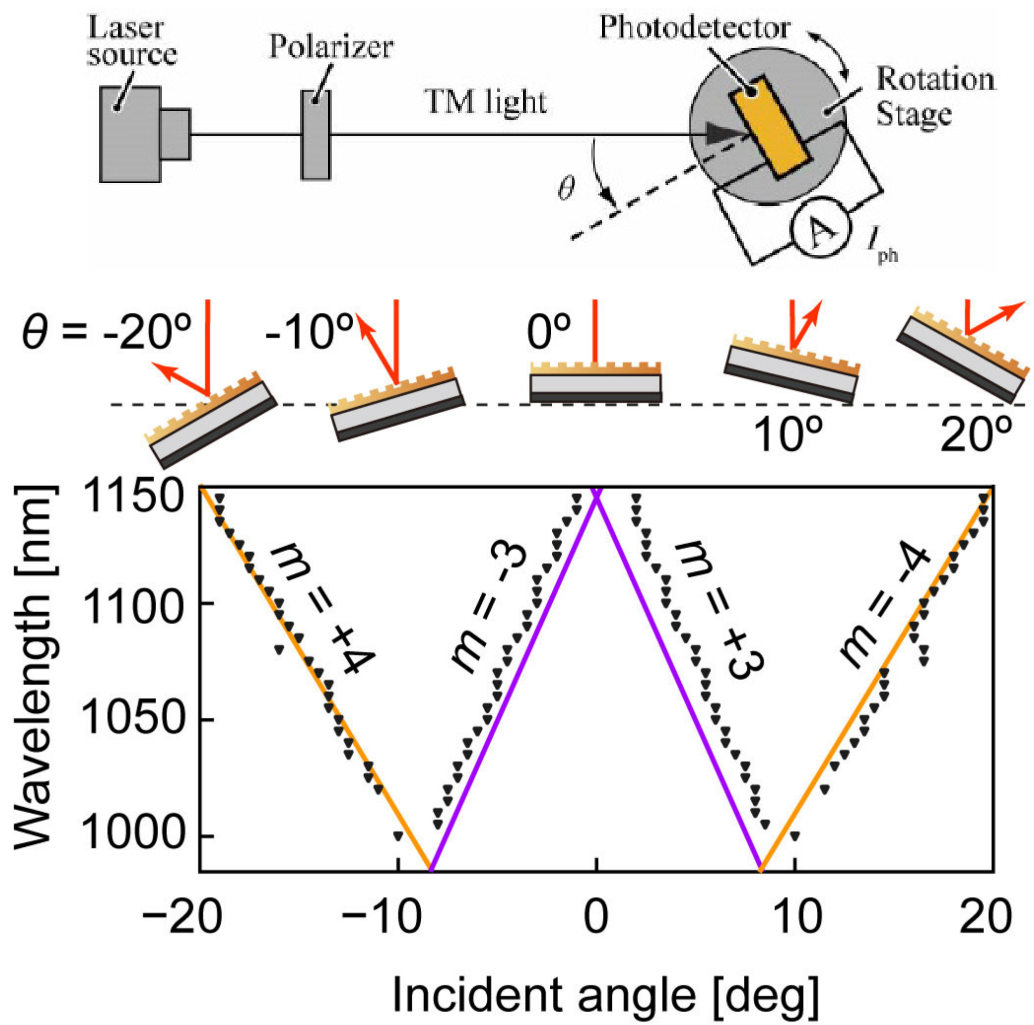




Supported by two zigzag root to decrease the stiffness

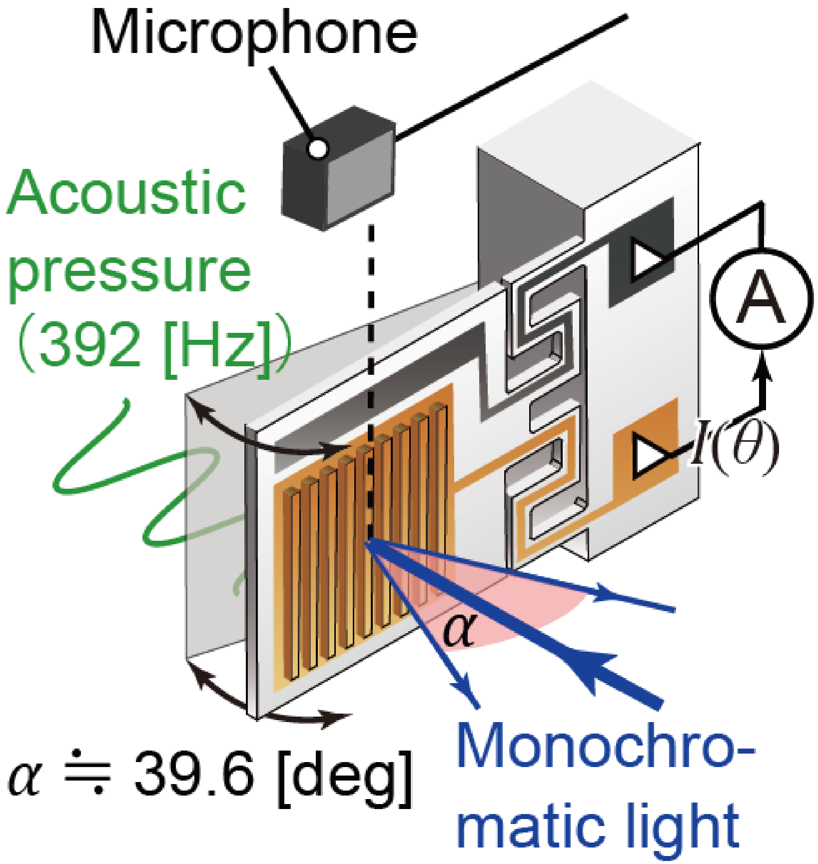
Static Plasmonic Photocurrent Response



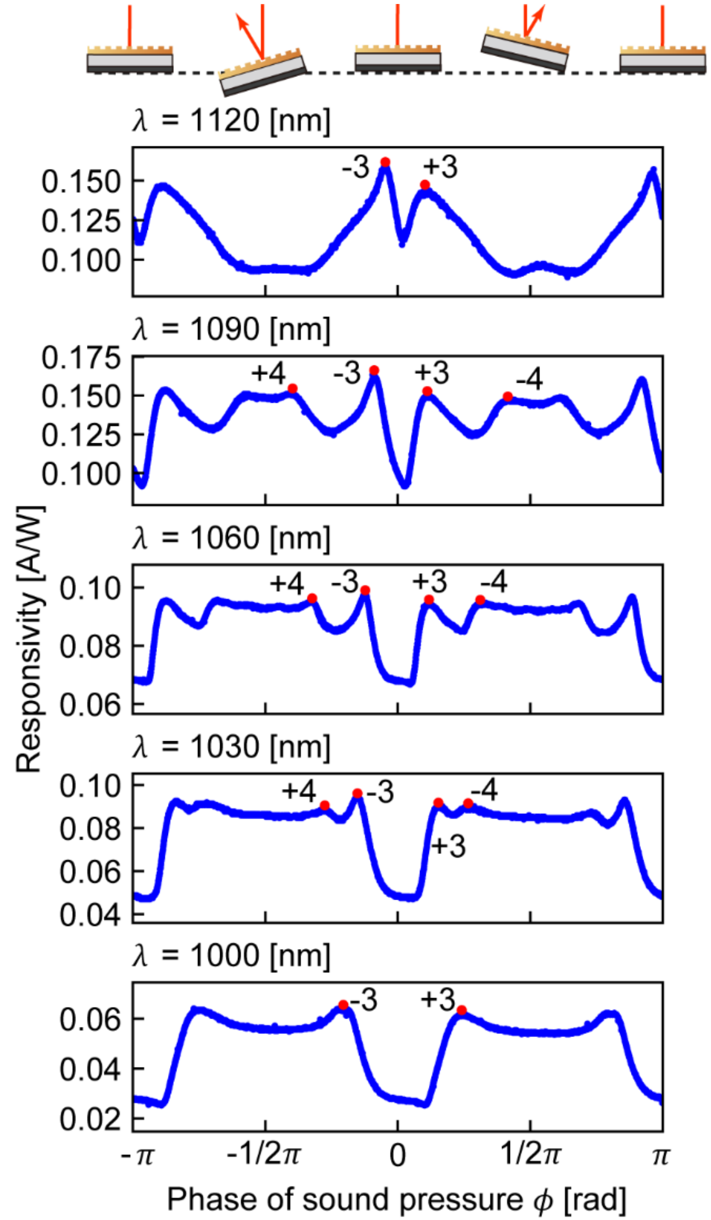


SPR successfully detected.

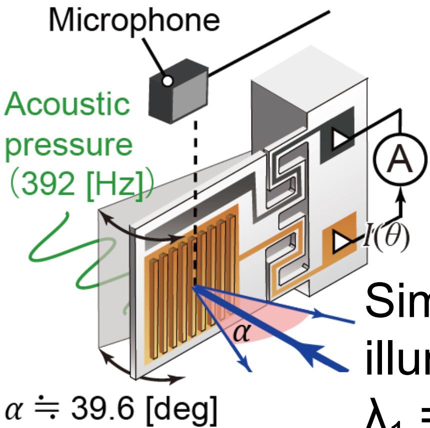
Dynamic Plasmonic Photocurrent Response



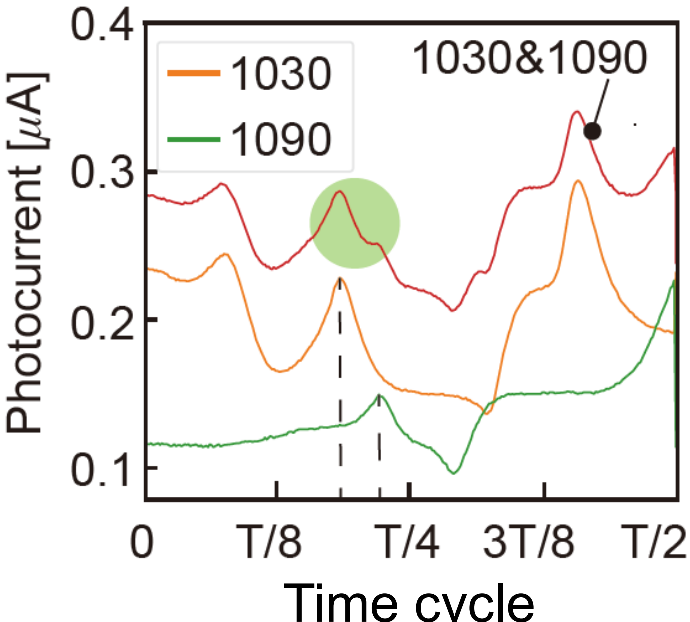
Dynamic modulation
by acoustic pressure



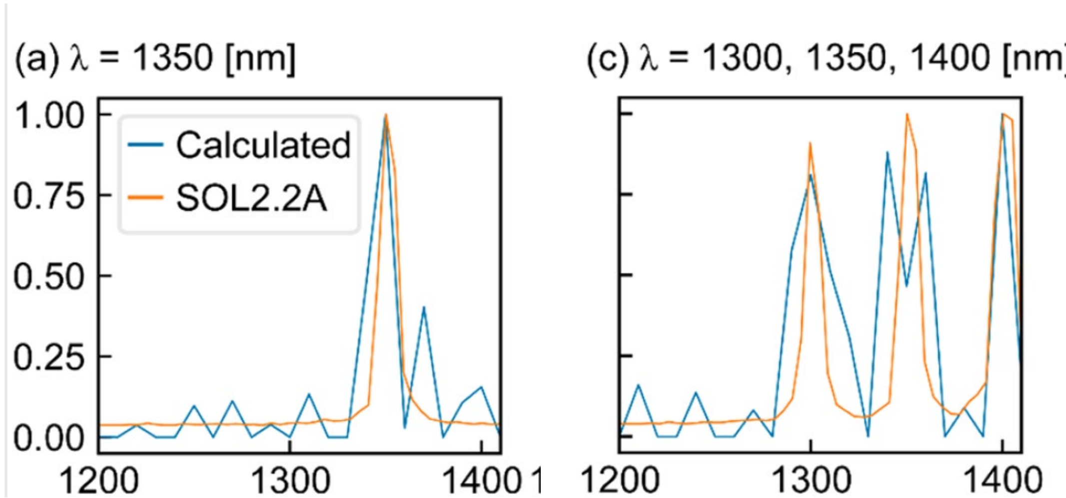
Spectroscopy on chip

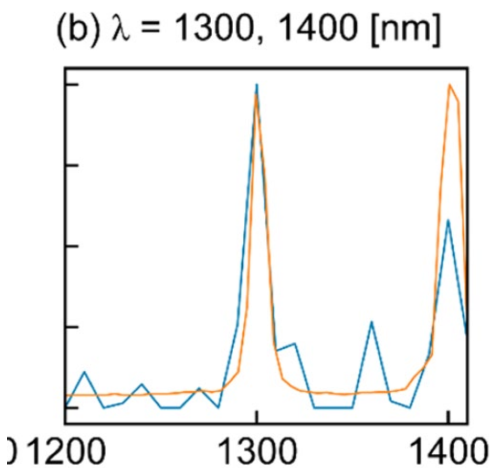


Simultaneous illumi.
λ_1 =1030nm
λ_2 =1090nm



Superposition of two wavelength
Responses confirmed





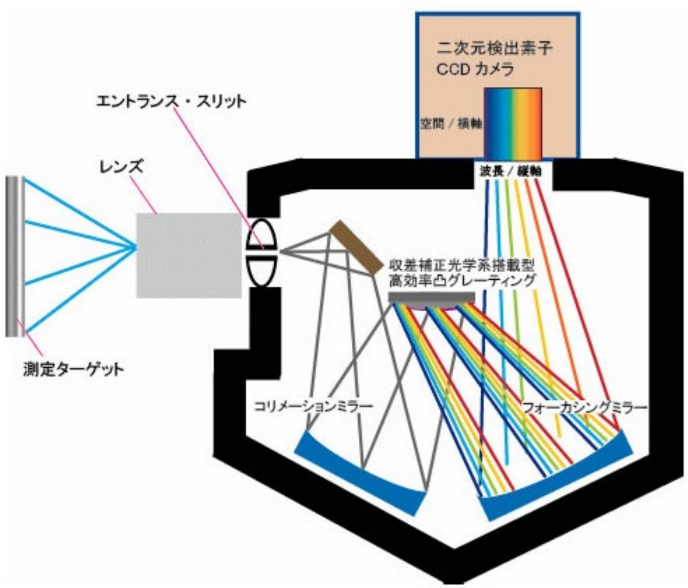
Orange : commercial spectrometer
Blue : MEMS spectrometer

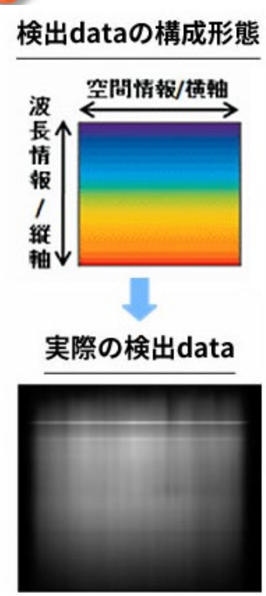
Ability of spectroscopy confirmed.

M. Oshita, ACS Photonics, vol. 7, no. 3, pp. 673 (2020)

There have already been a spectral camera.

Google car



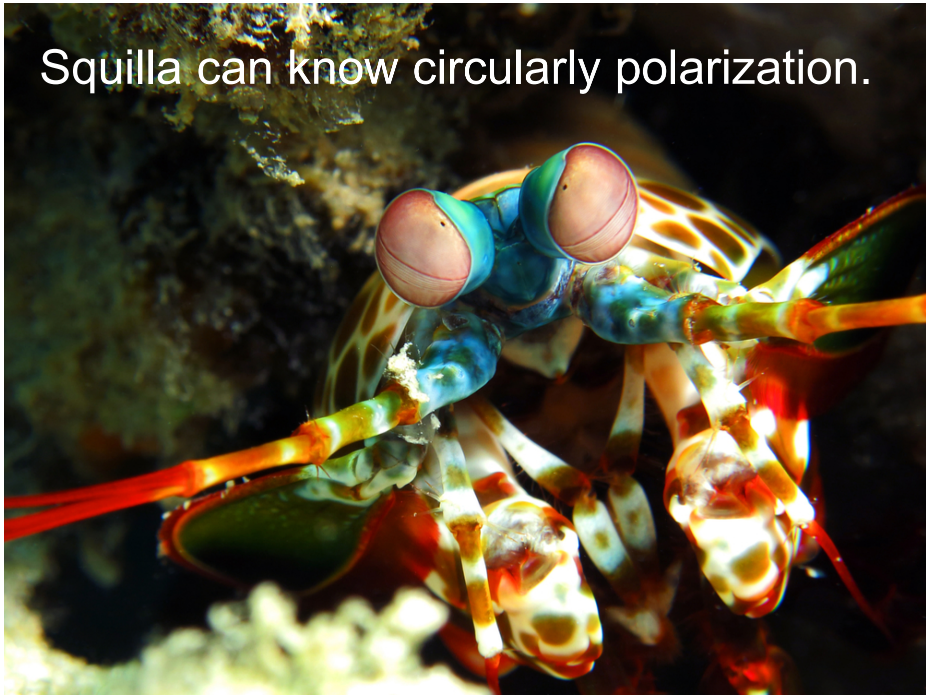


Hyper spectral camera
～10M JPY https://www.argocorp.com

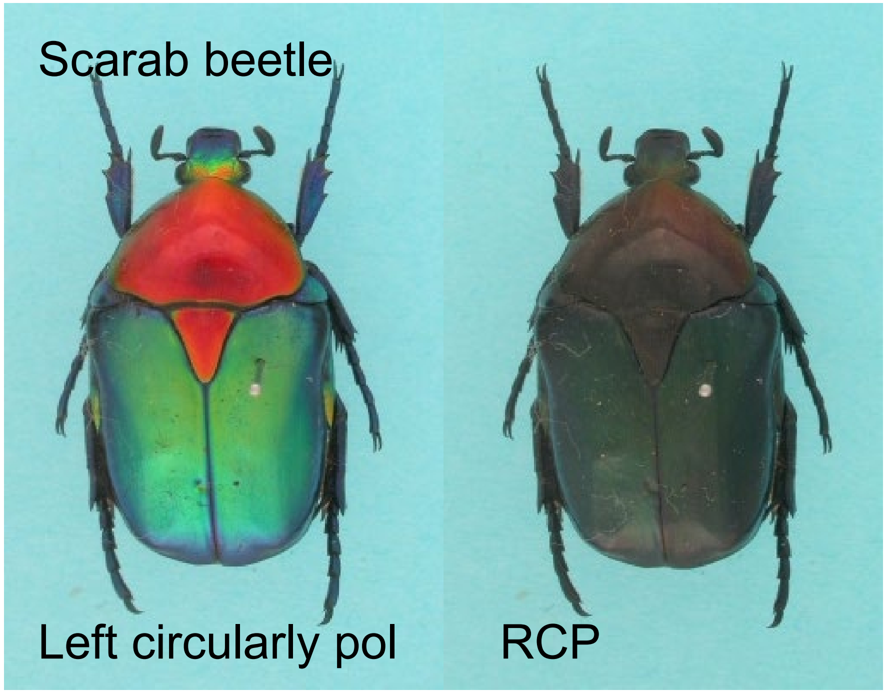
- Google car
- Not applicable for usual robot vision
- Compact & cost effectiveness will provide huge impact.

Contents

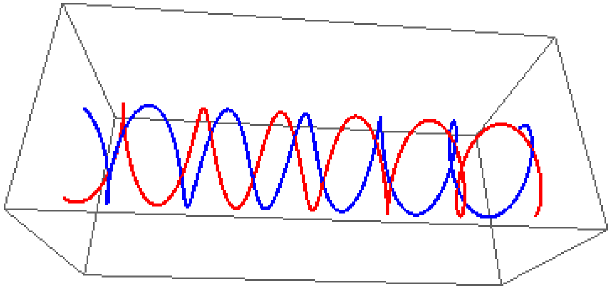
There are some species to use polarization.

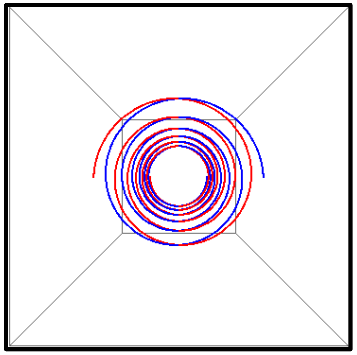


https://wired.jp/2008/04/01/



http://www.op.titech.ac.jp/lab/Take-Ishi/html/ki/hg/et/sb/goldbug/goldbug.html



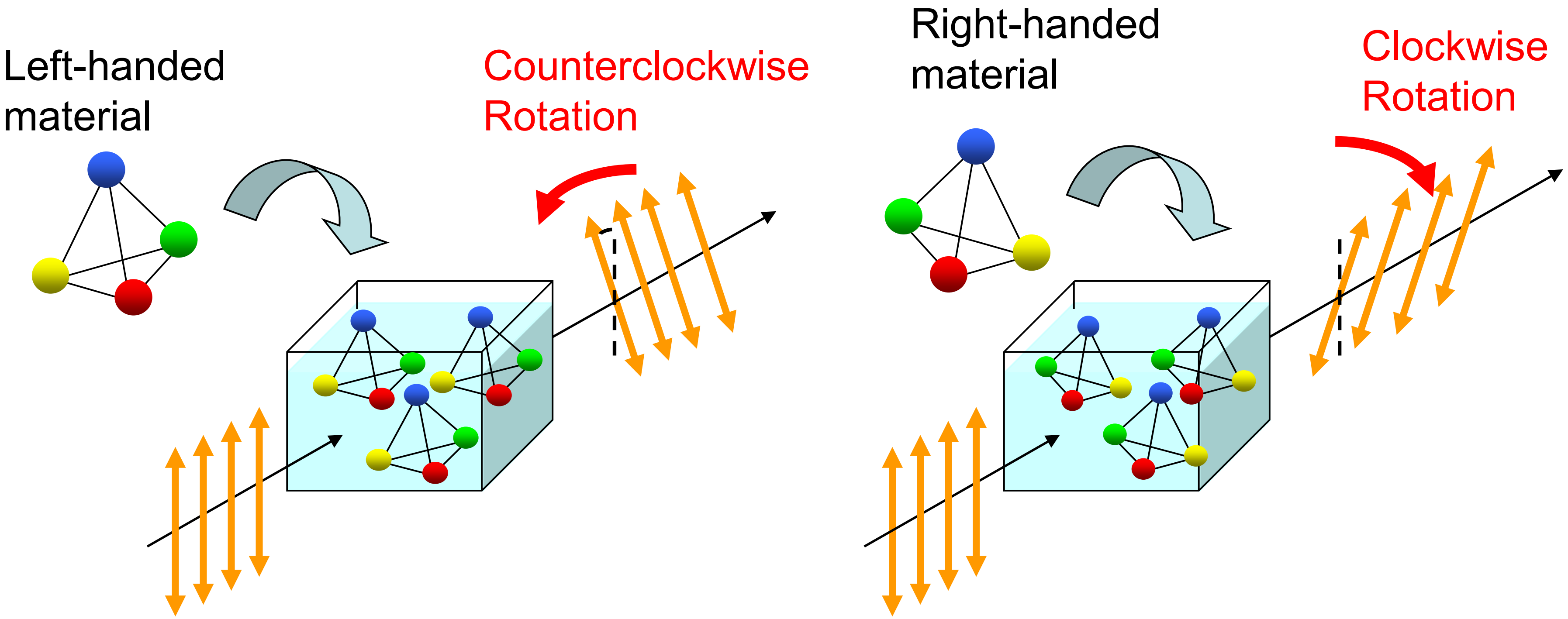


Red : RCP
Blue : LCP

Living things greedily use natural mechanisms on a process of evolution, which is often unimaginable for us human.
Studying nature may be important to find a new engineering frontier.

Polarization modulation by chiral media

■ Chiral media: no symmetry plane, optical activity



■Same transmittance, inverted polarization modulation

■Chiral switching gives polarization polarity change

Right circularly polarized⇔Left circularly polarized

■Application to THz polarization modulator

3D chiral switching

- 3D: Optical activity is strong
- Symmetry: Spectral shape is symmetry

- We employed **MEMS (Micro Electro Mechanical Systems)** technology.

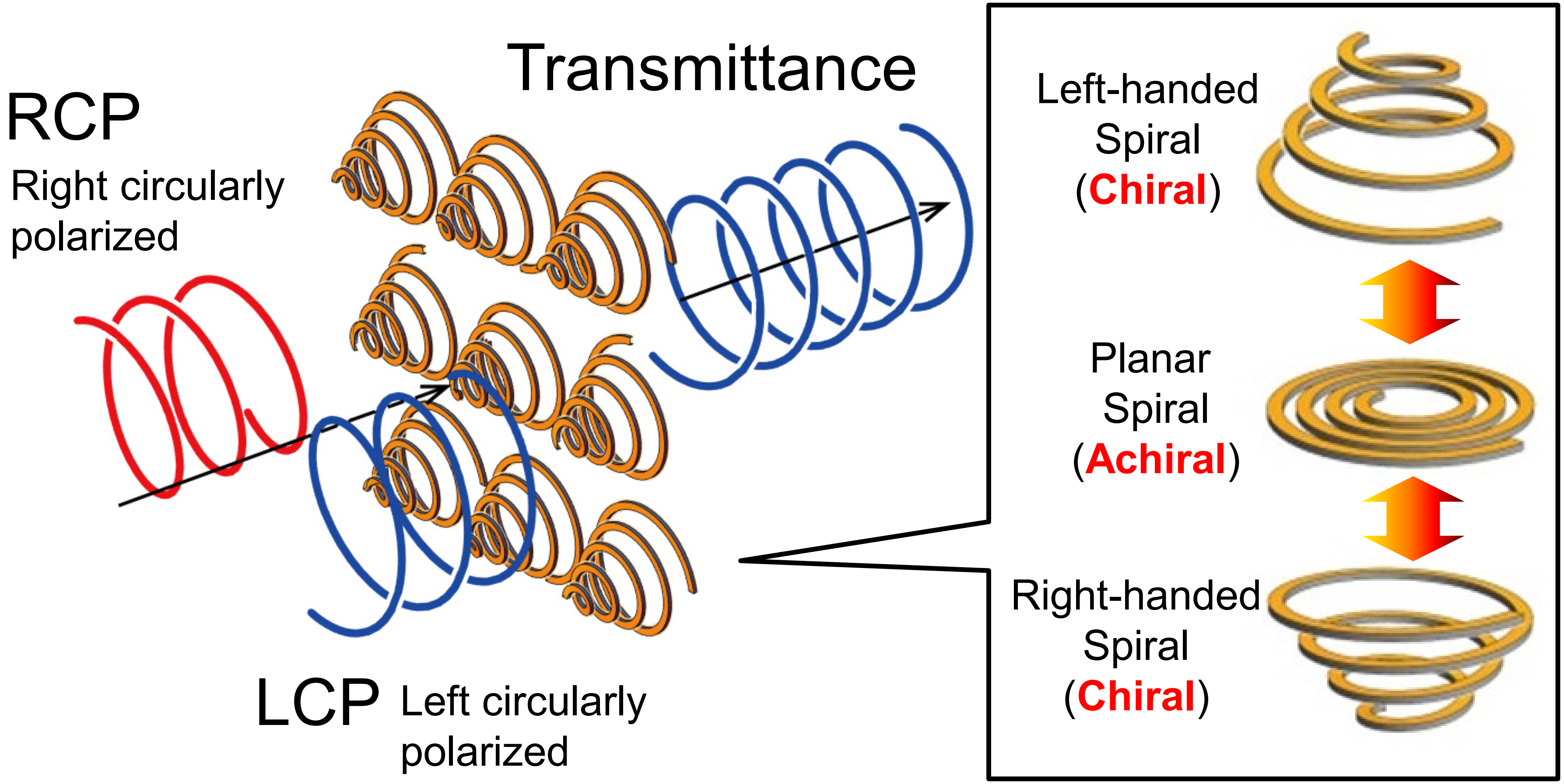
Right-handed Spiral (**Chiral**)

Planar Spiral (**Achiral**)

Left-handed Spiral (**Chiral**)

3D symmetrical chiral switching by the spiral

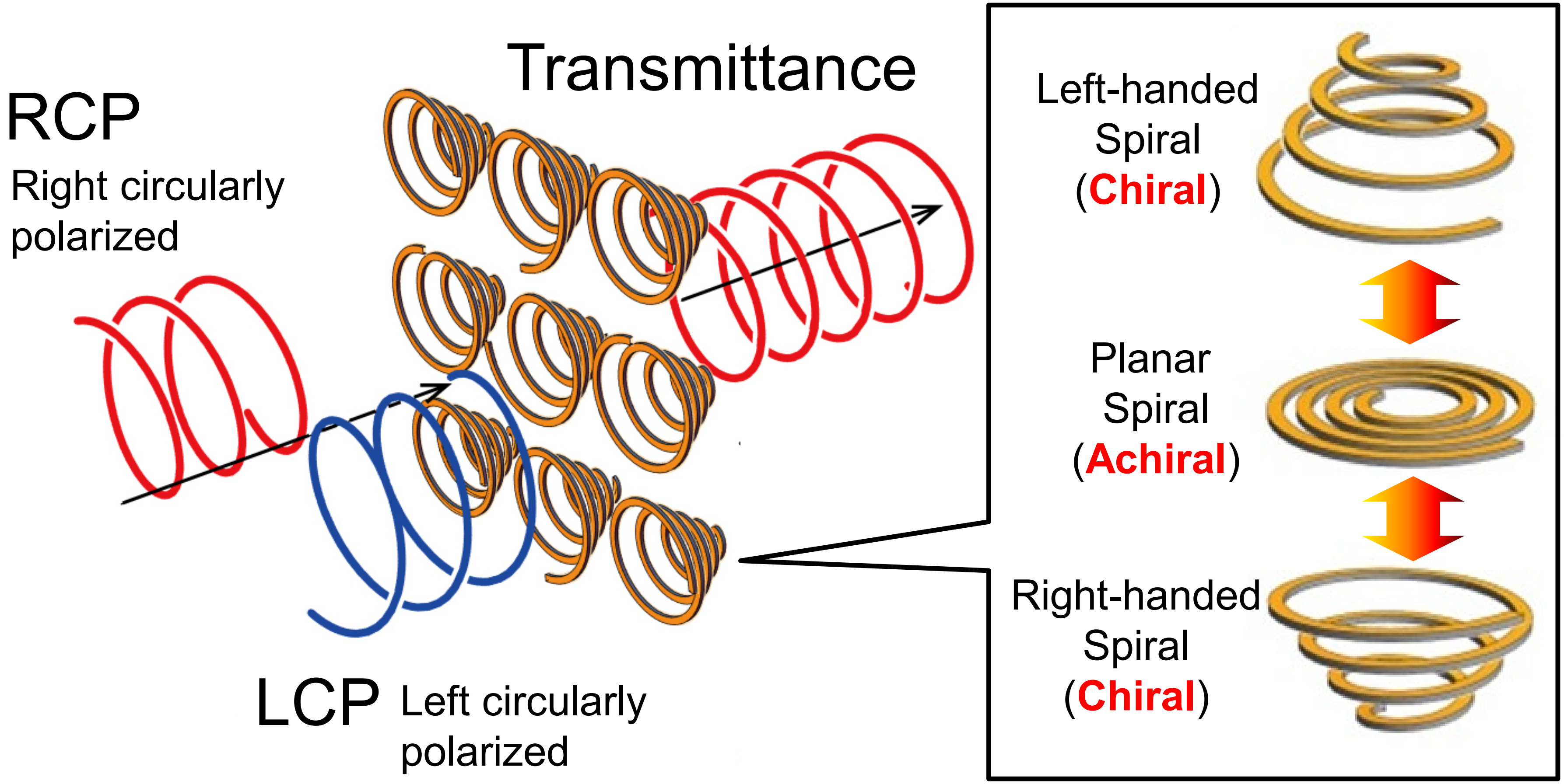
Bi-directional deformation



T. Kan, *et. al.*, *Nature communications,* vol. 6, art. no. 8422, 2015.

Micro spiral deformation
Tuning CP transmittance → Dynamic control of polarization

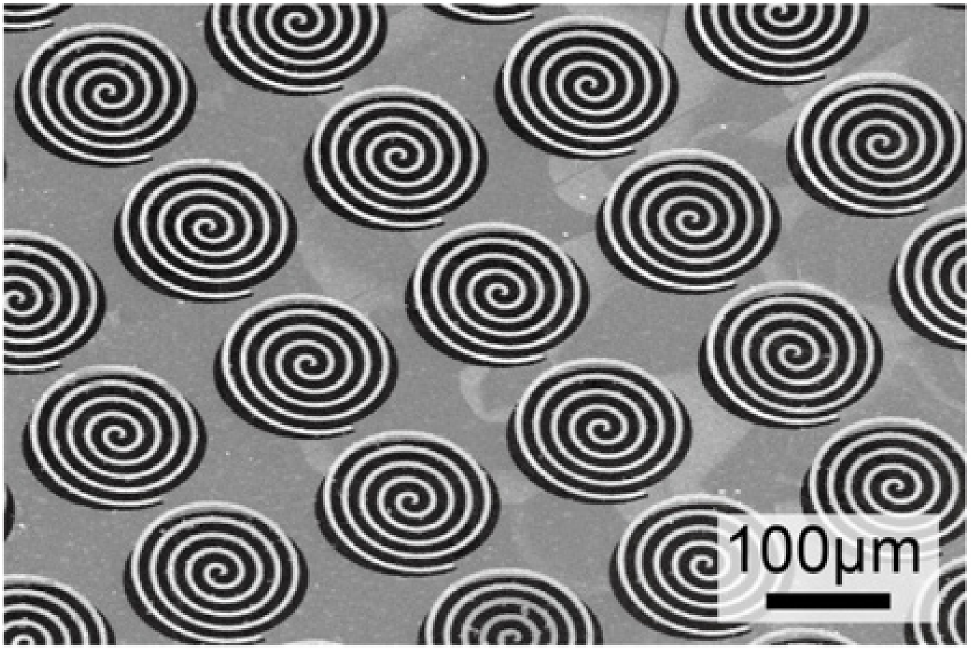
Bi-directional deformation

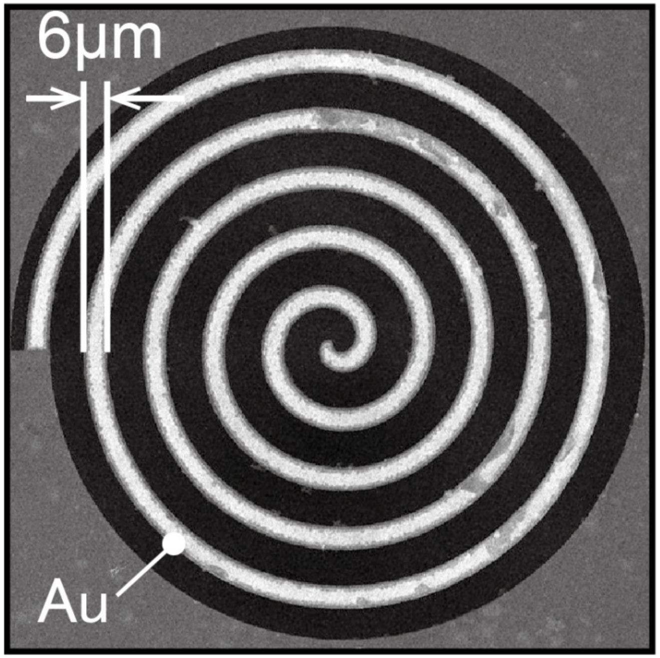


T. Kan, *et. al.*, *Nature communications,* vol. 6, art. no. 8422, 2015.

Micro spiral deformation
Tuning CP transmittance → Dynamic control of polarization

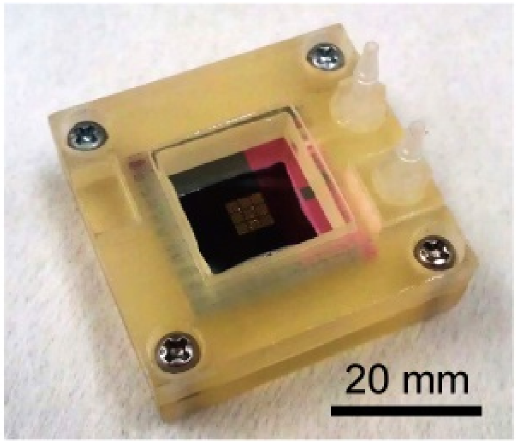
Fabricated MEMS metamaterial

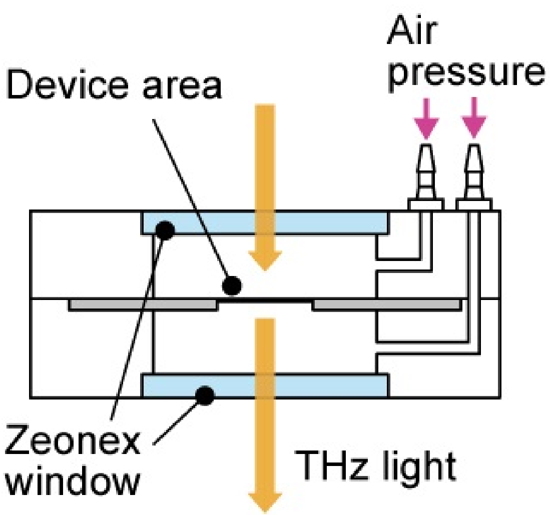


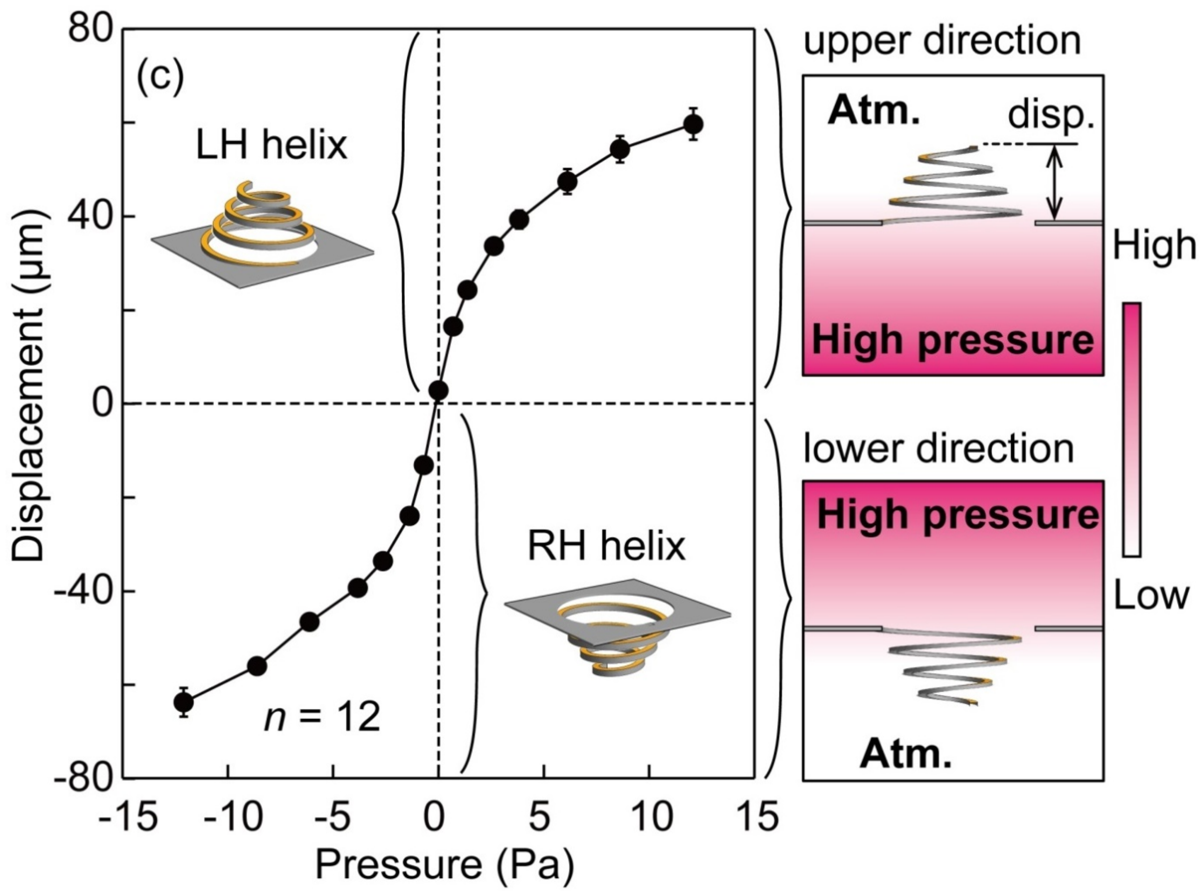


Frequency ~1 THz (λ~300 µm)
Diameter: 150 µm
Spiral shape: Archimedean spirals (5-turns)
Beam width: 6 µm, Thickness: 345 nm
Array pitch 170 µm

Pneumatic force for mechanical deformation

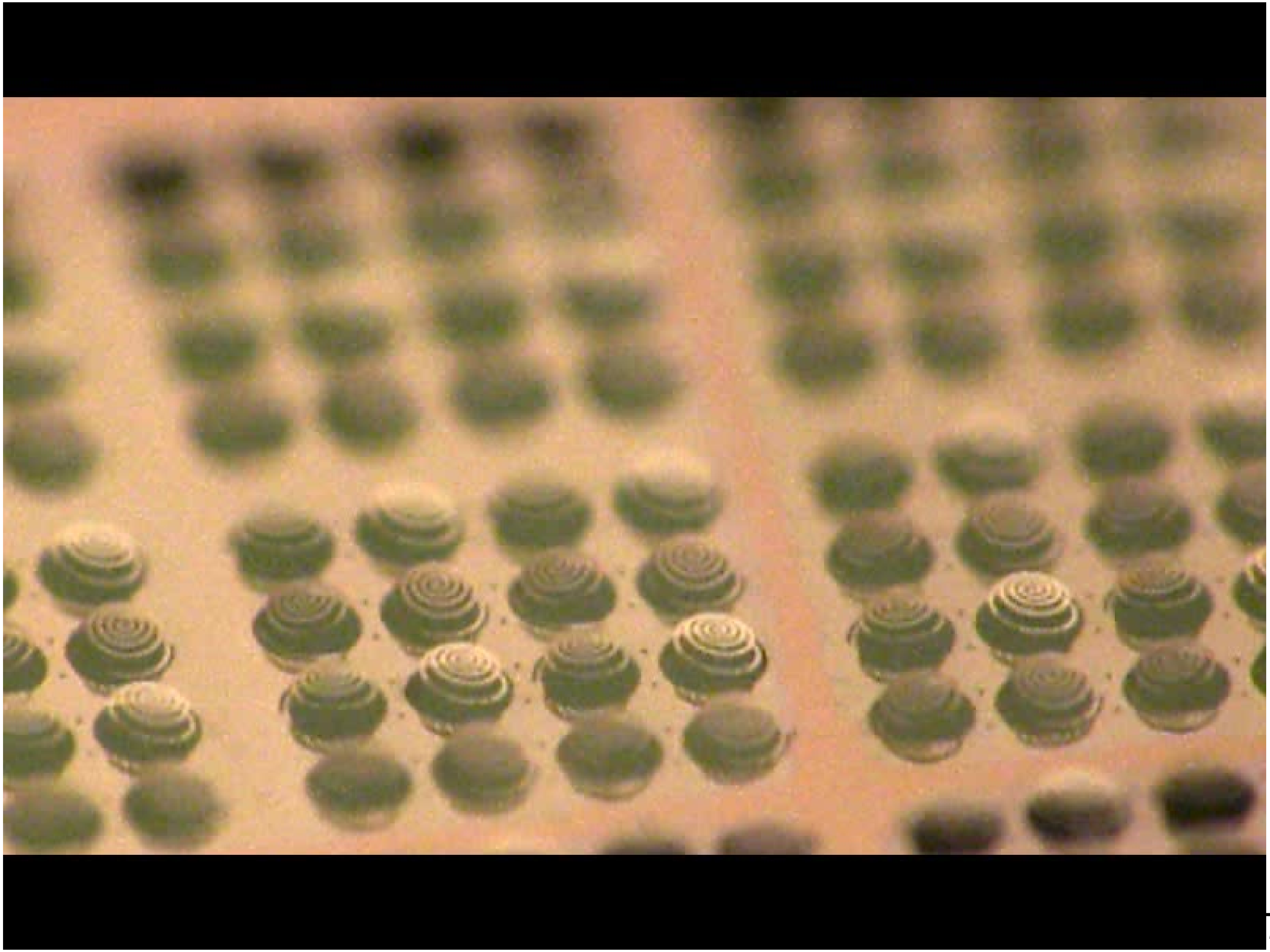






- ■N_2 pneumatic force
- ■Symmetrical deformation

Deforming spirals by pneumatic force



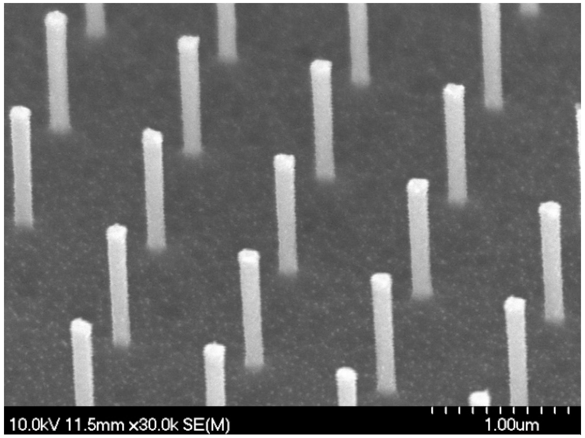
Summary and outlook

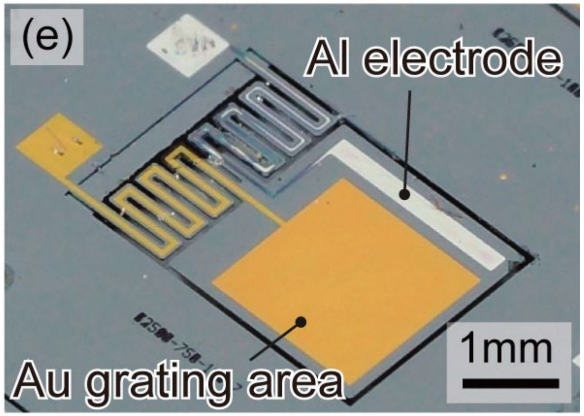
Learning vision systems of living things

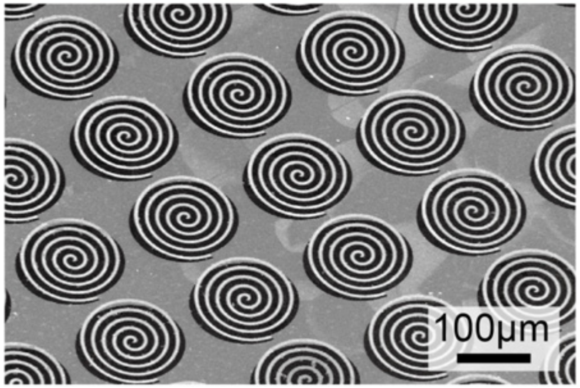
Going beyond them to broaden frontiers of intelligent mehcanics and robotics

Software and hardware are two wheels of the same car.
We cannot calculate without good data.

Even in AI hype era, study on a hardware is still required to provide highly competitive technology.







Assignment

- Pick up a vision/camera device, which recently became popular.
  (ex.) Go pro camera, 360degree camera, smart phone camera
- Discuss their impact on your life or on the world.
- State your idea how to evolve such a device.
- Summarize above points in a short essay with length ~500 words.